

First Full Publication Public Review Draft

BSR/ASHRAE/ICC Standard 240P

Quantification of Life Cycle Greenhouse Gas Emissions

First Full Public Review (February 2024)

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ASHRAE, 180 Technology Pkwy NW, Peachtree Corners, GA 30092



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FOREWORD

ASHRAE/ICC Standard 240P provides minimum requirement for quantifying both embodied and operational greenhouse gas emissions over the life-cycle of buildings, building systems, building equipment and their sites.

This standard has two paths. The first path is as-designed and the second path is as-built. In some sections both paths are allowed or may be combined based on the requirements of the authority having jurisdiction (AHJ). In addition, the committee has determined that the reference study period shall be 60 years unless otherwise required by the AHJ. The standard does include data for a 20-year reference study period and 100-year reference study period for ease of use.

This is the second public review of the standard. The first public review was an advisory public review which resulted in over 400 comments. The committee would like to thank all the commenters for their comments. During the last few months, the committee has reviewed all the comments and based on that review made substantial changes to the draft. These changes have resulted in reworking of whole sections and revising the title, purpose and scope.

ANSI/ASHRAE standards are reviewed every five years to determine if updating is needed. Due to the ever-evolving nature of whole-life carbon reduction, the committee will likely place the document on continuous maintenance once published. This will allow the document to be utilized and updated at a more rapid pace in the future.

Quantification of Life Cycle Greenhouse Gas Emissions of Buildings

1. Purpose: The purpose of this standard is to provide a methodology to quantify and document greenhouse gas emissions associated with buildings, building systems, and building equipment and their sites over their life cycle.

2. Scope

2.1 This standard provides minimum requirements for the quantification of embodied and operational greenhouse gas emissions associated with buildings, and their sites.

2.2 This standard provides minimum requirements for documentation of life cycle greenhouse gas emissions.

2.3 This standard does not set benchmarks or establish levels of building performance.

2.4 This standard shall not be used to circumvent any safety, health, or environmental requirements.

3.1 Definitions

adopting authority: the governmental unit, agency or agent that adopts this standard.

alteration: a replacement or addition to a building or its systems and equipment. Routine maintenance, repair, or a change in the building's use classification or category does not constitute an alteration.

as-designed: the state that precedes the *as-built* state of a *building* wherein the actual installed conditions, equipment, and systems are not yet completed or installed.

as-built: the state that represents the actual installed conditions, equipment, and systems of a *building*.

as-built building information model (BIM): building information model developed after the construction of a building or portion of a building that captures changes to materials used (type, size, location, etc.) as compared to those captured in the design documents.

authority having jurisdiction (AHJ): the governmental unit, agency, agent or code official responsible for enforcing this standard.

bill of materials: A comprehensive list of the quantity, cost, and mass of individual building products, assemblies and materials as delivered to the project site for permanent installation in the building project. Assemblies may be documented based on their constituent components.

biomass: organic material that comes from living organisms.

biogenic: produced in natural processes by living organisms, but not fossilized or derived from fossil resources.

building. Any *structure* used or intended for supporting or sheltering any use or occupancy.

building project: see *project*.

bulk fuel: an *imported energy* source that is stored at the building site and combusted or otherwise chemically converted on-site for the purpose of energy services, heating, or other amenities, including propane, fuel oil and other petroleum products, hydrogen, coal, wood, and other biofuel or *biomass* products.

carbonation: a chemical process that generates carbonates, bicarbonates, and carbonic acid.

carbon dioxide (CO₂): a naturally occurring gas, CO₂ is also a by-product of burning *fossil fuels* (such as oil, gas and coal), of burning *biomass*, of land-use changes (LUC) and of industrial processes (e.g., cement production). It is the reference gas against which other GHGs are measured and therefore has a *global warming potential (GWP)* of 1.

carbon dioxide equivalent (CO₂e) emission: the amount of *carbon dioxide* (CO₂) emissions that would cause the same integrated radiative forcing or temperature change, over a given time horizon, as an emitted amount of a greenhouse gas (GHG) or a mixture of GHGs.

carbon sequestration: process by which *carbon dioxide* is removed from the atmosphere; can be natural or artificial.

Conditioned space: an area, room or space that is enclosed within the *building thermal envelope* and is directly or indirectly heated or cooled. Spaces are indirectly heated or cooled where they communicate through openings with *conditioned spaces*, where they are separated from *conditioned spaces* by uninsulated walls, floors or ceilings, or where they contain uninsulated ducts, piping or other sources of heating or cooling.

construction documents: drawings and specifications used to construct, add to, or alter *buildings, systems, or equipment*, or portions thereof.

Construction stage: The life cycle stage composed of *Module A4* and *Module A5*.

construction waste: any substance, matter, or byproduct generated from the *building* construction or *demolition* process.

Cradle-to-gate: see *product stage*.

cradle-to-grave scope: a scope that includes the *product stage*, *construction stage*, *use stage*, and *end-of-life stage*.

decarbonization: the process of removing or reducing *greenhouse gas emissions*.

deconstruction: the controlled and intentional selective dismantlement of a *building* or *building elements* in a manner to preserve valuable components for the purposes of reuse, repurposing, and/or recycling.

demolition: the controlled and intentional action or process of dismantling, razing, destroying, or wrecking of a *building* or *building elements*.

Design for deconstruction: designing in such a way that the resources used can be economically recovered and reused.

disposal management: activity that includes on-site transport and energy and water use related to waste processing at the disposal location, including environmental loads from energy recovery processes.

District energy system: a system that supplies cooling or heating energy to multiple buildings, consisting of one or more plants producing hot water, steam and/or chilled water and distribution systems of pipes to convey hot water, steam or chilled water to the buildings.

electric vehicle: an automotive-type vehicle for on-road use, such as passenger automobiles, buses, trucks, vans, neighborhood *electric vehicles*, plug-in hybrid vehicles, and electric motorcycles, primarily powered by an electric motor that draws current from a building electrical service or another source of electric current.

electric vehicle supply equipment: equipment for plug-in electricity transfer including the ungrounded, grounded and equipment grounding conductors, and the *electric vehicle* connectors, attachment plugs, personal protection system and all other fittings, devices, power outlets or apparatus installed specifically for the purpose of transferring energy between the premises wiring and an *electric vehicle*.

embodied greenhouse gas emissions: the GHG emissions associated with the extraction, production, transportation, construction or assembly, *deconstruction*, *demolition*, end-of-life, and use of building or building site materials.

emissions factor: a representative value that attempts to relate the quantity of a pollutant released to the atmosphere with an activity associated with the release of that pollutant. These factors are usually expressed as the weight of pollutant divided by a unit weight, volume, distance, or duration of the activity emitting the pollutant.

End-of-life stage: *The life-cycle stage that occurs after the end of the reference study period and is comprised of Module C1, Module C2, Module C3, and Module C4.*

Energy code: a section, chapter or subset of a building code containing written legal requirements that govern the energy efficiency design and construction of residential or commercial *structures*.

Environmental product declaration (EPD): A Quantified environmental information on the life cycle of a product.

Equipment: devices for energy conversion or transformation, space heating, space cooling, ventilation, humidification, dehumidification, electric power, lighting, transportation, refrigeration, cooking, or service water heating, or other process loads at a building site.

Existing building: A *building* or portion thereof that was previously occupied or approved for occupancy by the authority having jurisdiction.

Exported energy: Energy that is delivered from within the asset boundary for use outside the asset boundary. *Exported energy* may result from conversions of renewable energy harvested within the asset boundary or may result from conversions of non-renewable energy that is delivered to the asset and measured as energy consumed by the asset.

Exterior wall: a wall, bearing or nonbearing, that is used as an enclosing wall for a building, other than a fire wall, and that has a slope of 60 degrees (1.05 rad) or greater with the horizontal plane.

Fossil fuels: non-renewable energy sources that formed from the decomposition of buried, carbon-based organisms from the geological past, such as coal, natural gas, propane, fuel oil and other petroleum products.

Fugitive emissions: the emission, leakage or discharge of gases or vapors from pressure-containing equipment or facilities and components inside buildings such as valves, piping flanges, pumps, storage tanks, valves, compressors, etc.

greenhouse gas (GHG): Gaseous constituents of the atmosphere, both natural and *anthropogenic*, that absorb and emit radiation at specific wavelengths within the spectrum of terrestrial radiation emitted by the Earth's surface, the atmosphere itself and by clouds. This property causes the greenhouse effect. Water vapor (H₂O), carbon dioxide (CO₂), nitrous oxide (N₂O), methane (CH₄) and ozone (O₃) are the primary GHGs in the Earth's atmosphere.

Greenhouse gas emissions: The release of *greenhouse gases* into the atmosphere that absorb and emit radiation at specific wavelengths within the range of the electromagnetic spectrum that radiation is emitted by the Earth's surface, the atmosphere and clouds.

greenhouse gas emissions, combustion: *greenhouse gas* emissions associated with the burning of a solid or liquid or gaseous fuel, either within the *system boundary* or to generate electricity, steam, hot water or chilled water that is generated outside the *system boundary* and used within the *system boundary*.

Greenhouse gas emissions, precombustion: *greenhouse gas* emissions associated with fuel extraction, processing, and transport, including *fugitive emissions*, prior to combustion within the *system boundary* or to generate electricity or thermal energy used within the *system boundary*.

Global warming potential (GWP): an index developed to provide a simplified means of describing the relative ability of a chemical compound to affect radiative forcing, if emitted to the atmosphere, over its lifetime in the atmosphere, and thereby to affect the global climate. The GWP is defined on a mass basis relative to *carbon*

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dioxide. The GWP for a compound must be calculated up to a particular integrated time horizon, for example, 20, 100, or 500 years.

Imported electricity: electricity that is generated outside the *system boundary* and is delivered for use within the *system boundary*.

Imported energy: an energy source that is delivered for use within the *system boundary*. Energy that is transported from outside the *system boundary* for use within *the system boundary* including *imported electricity*. Imported energy does not include naturally occurring renewable energy, such as wind, solar or geothermal, that is harvested and converted within the system boundary for on-site use or for export.

Industrial gases: the gaseous materials that are manufactured for use in industry and include but are not limited to nitrogen, oxygen, carbon dioxide, argon, hydrogen, helium and acetylene.

Industry-average EPD: *Environmental Product Declaration (EPD)* that declares average values for products of multiple manufacturers in a clearly defined sector and/or geographical area.

Land use change: process by which human activities transforms or modifies natural landscape.

Licensed design professional: individual responsible for the design and preparation of architectural or engineering contract documents. An architect or engineer licensed to practice in accordance with applicable state licensing laws.

Life-cycle assessment (LCA): the process of evaluating a *project* from *cradle-to-grave* and assigning a value or assessment of its cumulative and ultimate social, environmental and economic costs, benefits, and impacts.

material Inventory: the quantified list of the materials, products, and waste inputs and outputs over the lifecycle of the project based on scenarios applied to the *bill of materials*.

Module A1: The life cycle module corresponding to all processes and materials associated with harvesting, extraction, collection, and further processing of raw materials.

Module A2: The life cycle module corresponding to the transport of raw materials to a product manufacturing facility or to multiple manufacturing facilities.

Module A3: The life cycle module corresponding to the processes and materials required for the fabrication and production of a product prior to Module A4.

Module A4: The life-cycle module corresponding to the transportation of all materials and products included in the *bill of materials*.

Module A5: The life cycle module corresponding to the processes and materials required for the construction of the *project* on a *site* prior to the issuance of a certificate of occupancy.

Module B1: The life cycle module corresponding to the non-energy related emissions that occur during the life of the building associated with normal conditions of use of the *products* and *components* installed in the *project*, including greenhouse gas emissions from off-gassing materials, processes such as *carbonation*, and *fugitive emissions*.

Module B2: The life-cycle module corresponding to all processes and materials used for planned maintenance required to sustain the functions of the building and building systems, including the production, transportation, and end-of- life treatment of materials used in the maintenance processes.

Module B3: The life-cycle module corresponding to all processes and materials required to sustain the functions of the building that are not included in Module B2, including production and transport of materials and components, waste management and end-of-life treatment of materials and components removed from the building, and production, transport and disposal of ancillary materials used during the repair.

Module B4: The life-cycle module corresponding to all processes, materials, and components used in replacing components in their entirety at the end of their service life, including production and transportation of new materials and components, waste management and end-of-life treatment of materials and components removed from the building, and production, transport and disposal of ancillary materials used during the replacement.

Module B5: The life-cycle module corresponding to all processes, materials, and components used during refurbishment or retrofit processes within the larger project, including production and transportation of new materials and components, waste management and end-of-life treatment of materials and components removed from the building, and production, transport and disposal of ancillary materials used during the refurbishment activity.

Module B6: The life-cycle module corresponding to operational energy use of the project.

Module B7: The life-cycle module corresponding to operational water use and its treatment.

Module C1: The life-cycle module corresponding to all on-site and off-site processes and inputs for the decommissioning, dismantling, deconstructing, or demolishing the project.

Module C2: The life-cycle module corresponding to all transportation between the site and the final sorting, disposal, or end-of-waste location, including transportation to and from intermediate storage or waste processing locations.

Module C3: The life-cycle module that includes all processes and material flows required for recycling, reusing, or recovering construction products, materials, construction elements, and debris to reach their end-of-waste state as recovered material reentering the market.

Module C4: The life-cycle module corresponding to all processes and material flows required for waste disposal and treatment—including neutralization, incineration, landfilling, and disposal management.

Module D: The module comprised of Module D1 and Module D2, which quantifies benefits and loads outside of the system boundary.

Module D1: The life-cycle module corresponding to all environmental loads and benefits from future substitution of resources due to reused products, recycled materials, secondary fuels, and recovered energy leaving a project for use in a subsequent product system as material.

Module D2: The life-cycle module corresponding to all environmental loads and benefits for recovered and exported energy used to meet the energy demand outside of the project.

operational carbon emissions: the greenhouse gas emissions associated with the operation of an asset (i.e., building) during the use stage of the asset.

product category rule (PCR): product-specific guidelines that determine the requirements for conducting life cycle assessment studies and reporting findings through EPDs.

product-specific EPD: *Environmental Product Declaration (EPD)* developed by a single-manufacturer that provides data on a single product.

product stage: The life cycle stage comprised of Module A1, Module A2 and Module A3.

project: *a construction work that includes a building, system, or site.*

record documents: drawings and other documents that record the conditions of the project as constructed. These include any refinements of the *construction documents* or bid documents.

reference study period (RSP): the period over which the time-dependent characteristics of the object of assessment are analyzed.

Informative note: The *RSP* is not an assumption of the exact service life of a particular product, system or assembly but is intended to represent a consistent service life for building types covered by this standard that is predictable and relevant for carbon accounting of buildings.

site: The area of land adjacent to and intended to support the use of a *building*. A site shall be contiguous or separated only by public rights-of-way.

structure: that which is built or constructed.

substructure: a portion of a *structure* that is built at or below grade.

superstructure: a portion of a *structure* that is built above grade.

system boundary: the physical, geographical, and temporal scope of the assessment, including life cycle stages, *building* elements, processes, flows, and activities.

Use stage: The life cycle stage that occurs from issuance of a certificate of occupancy to the end of the *reference study period* and is comprised of *Module B1, Module B2, Module B3, Module B4, Module B5, Module B6, and Module B7*.

3.2 Abbreviations and Acronyms

AHJ	authority having jurisdiction
BIM	building information modeling
CO ₂ e	carbon dioxide equivalent emissions
EPD	environmental product declaration
GHG	greenhouse gas emissions
GWP	global warming potential
LCA	life cycle analysis
MEP	mechanical, electrical, plumbing
PCR	product category rule

4. COMPLIANCE, ADMINISTRATION, AND ENFORCEMENT

4.1 Compliance. Compliance with this standard shall be based on the *System Boundary* determined in accordance with Section 5. As applicable to the *System Boundary*, quantification of *embodied greenhouse gas emissions* in accordance with Section 6, quantification of *operational greenhouse gas emissions* in accordance with Section 7, evaluation of loads and benefits beyond the system boundary in accordance with Section 8 and documentation of the analysis and results in accordance with Section 9 are required.

4.2 Assessment Type. Compliance with Sections 6, 7, and 8 requires completion of either an *as-designed* assessment or an *as-built* assessment. Determination of the assessment type used for compliance with each Section shall meet the requirements specified in the relevant section and may be different among sections.

4.3 Compliance Documentation. *Construction documents or record documents*, as applicable, shall show all the pertinent data and features of the *building, systems, equipment* and *sites* in sufficient detail to permit a determination of compliance by the *authority having jurisdiction (AHJ)* and to indicate compliance with the requirements of this standard.

4.4 Administrative Requirements. Administrative requirements relating to permit requirements, enforcement by the *AHJ*, locally adopted standards, interpretations, claims of exemption, and rights of appeal are to be specified by the *AHJ*.

4.5 Other Laws. The provisions of this standard shall not be deemed to nullify any provisions of local, state, or federal law. Where there is a conflict between a requirement of this standard and such other law affecting this calculation methodology, precedence shall be determined by the *adopting authority (AHJ)*.

4.6 Normative Appendices. The normative appendices to this standard are mandatory requirements of this standard, which, for reasons of convenience, are placed apart from all other normative requirements.

4.7 Informative Appendices and Notes. The informative appendices and informative notes contain additional information and are not mandatory or part of this standard.

4.8 Referenced Standards. The normative referenced standards herein and listed in Section 10 shall be considered part of the requirements of this standard to the prescribed extent of such reference. Where differences occur between the provisions of this standard and referenced standards, the provisions of this standard shall apply. Informative references are identified in Informative Appendix D to acknowledge sources and are not part of this standard.

- 5. System Boundary.** An evaluation of life cycle *GHG emissions* performed in compliance with this standard shall be based on a *system boundary* which includes:
- a. A spatial boundary as determined in accordance with Section 5.1.
 - b. *Building* elements identified in Section 5.2.
 - c. Effects outside the spatial boundary identified in accordance with Section 5.3
 - d. *Life-cycle assessment* modules as specified in Section 5.4
 - e. The *reference study period* as determined in accordance with Section 5.5.

Figure 5 illustrates the life cycle stages that shall be included in the *system boundary*.

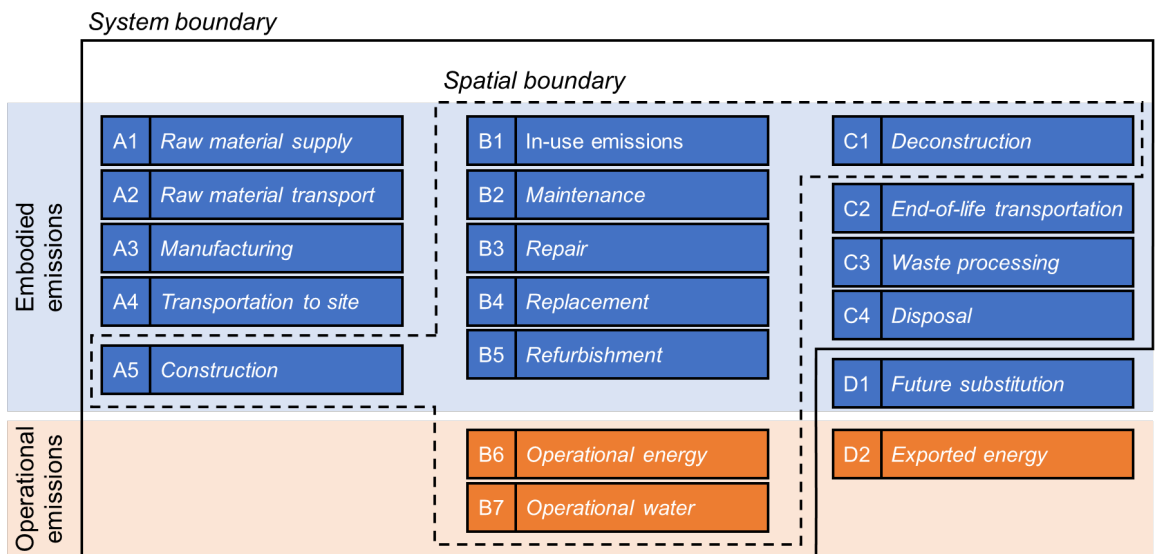


Figure 5: Illustration of life cycle stages included in the *system boundary*

5.1. Spatial Boundary. The spatial boundary shall identify the physical limits of the construction work of the *project*. A spatial boundary shall be a *building* or a portion of a *building* and relevant portions of the *site*, in accordance with Table A1: OmniClass Building Elements. Infrastructure, *building* components, and *site* components that are non-contiguous (e.g., off-site renewable energy systems) shall not be included within the spatial boundary.

Informative note: It should be possible to circle the spatial boundary on construction drawings.

5.2. Building Elements. All the *building* elements listed in Normative Appendix A which are applicable to the spatial boundary identified in section 5.1 shall be included in the *system boundary* and addressed in the evaluation.

5.3. Effects outside the Spatial Boundary. Emissions effects which occur outside the spatial boundary due to construction within the spatial boundary shall be included in the system boundary. Examples include utility and non-utility energy supply systems and infrastructure, *building* components, and *site* components that serve multiple buildings such as *district energy systems* and renewable energy where contiguous with the spatial boundary.

5.4. Scope of Life Cycle Stages. The *system boundary* shall address all the following *life-cycle stages*:

- a. Modules A1 through A5,
- b. Modules B1, B4, B5, and B6,
- c. Module B7 where required by the AHJ, and
- d. Modules C1 through C4.

The *system boundary* determined within the scope of this standard shall not include any of the following:

- a. Quantification of on-site *biogenic carbon sequestration*,
- b. avoided *GHG emissions*,
- c. negative emissions,
- d. carbon capture activities,
- e. carbon offsets,
- f. carbon credits,
- g. *renewable energy* credits, or
- h. other environmental attribute crediting mechanisms.

5.5. Reference Study Period. The *reference study period* shall be 60 years, unless otherwise stated by the AHJ.

5.6. Use Cases. Reserved for future use.

6. Quantifying Embodied Greenhouse Gas Emissions

6.1 General

6.1.1 Scope

Section 6 specifies requirements for the *embodied greenhouse gas emissions* assessment for *as-designed* and *as-*

built projects across life cycle stages A-D. (See Section 5.4.)

6.2 Establishing a *Material Inventory*

The assessment shall include a *material inventory* that includes all the following within the *system boundary* in accordance with Section 5.2:

1. Every *building* element representing more than 1% of total project cost or mass, and
2. Every *building* element described in Section 6.2.1.

6.2.1 Required *Building Elements*

6.2.1.1 *Substructure.*

The *assessment* shall include all the following *building* elements:

- a. Foundations: Standard shallow and deep foundations, and special foundations
- b. Subgrade Enclosures: Retaining walls, *site* walls.
- c. Slabs-on-Grade: Standard and structural flatwork, slab trenches, pits and bases, and slab-on-grade supplementary components
- d. Water and Gas Mitigation: *Building* subdrainage

Exception to Section 6.2.2.1: *Substructure building* elements that in aggregate represent less than 5% of the total substructure cost or mass may be excluded.

6.2.1.2 *Superstructure.*

The *assessment* shall include all the following *building* elements:

- a. Gravity systems: columns, bearing walls, beams, joists, girders, floor plate, roof plate, topping slabs, ledgers, blocking, deck edge angles, and kickers.
- b. Lateral systems: bracing and frames, shear walls, diaphragms.
- c. Connections & Attachments: primary connections (framing to framing), secondary connections (sheathing and nailing), diaphragm connections, secondary reinforcement (drag bars, opening reinforcement, add bars).
- d. Stairs: Stringers, treads, and railing.

Exception to Section 6.2.1.2: *Superstructure building* elements that in aggregate represent less than 5% of the total *superstructure* cost or mass may be excluded.

6.2.1.3 *Enclosure.*

The assessment shall include all the following *building elements*:

- a. Exterior Vertical *Enclosures*: Walls, windows, doors and grilles, louvers and vents, wall appurtenances, wall specialties
- b. Exterior Horizontal *Enclosures*: Roofing, traffic bearing horizontal enclosures, horizontal openings, overhead exterior *enclosures*.

Exception to Section 6.2.1.3: *Enclosure building elements* that in aggregate represent less than 5% of the total enclosure cost or mass may be excluded.

6.2.1.4 Interior.

The assessment shall include all the following *building* elements:

- a. Interior Construction: Partitions, walls, windows, doors, raised floor construction, suspended ceiling construction.
- b. Interior Finishes: wall finishes, interior fabrications, flooring, ceiling finishes.

Exception to Section 6.2.1.4: Interior *building* elements that in aggregate represent less than 5% of the total interior cost or mass may be excluded.

6.2.1.5 MEP and Services.

The assessment shall include all the following *building* elements:

- a. Mechanical Equipment and their Distribution Systems: ducts and pipes serving mechanical equipment and their insulation, chillers, cooling towers, air handlers, heat pumps, boilers, chilled beams, VRF systems, heat exchangers, and air conditioners.
- b. Electrical: Light fixtures, power supply and distribution, and power management systems, including batteries, uninterrupted power supplies, switchgears, transformers, and power generation systems.
- c. Plumbing Equipment Systems: Pipes servicing equipment and fixtures and their ins.

Exception to Section 6.2.1.5: MEP and Services *building elements* that in aggregate represent less than 5% of the total MEP and Services cost or mass may be excluded.

6.2.1.6 Civil/Landscape/Site Material Quantity Calculations. Reserved for future use.

6.2.2 Material Quantity Data Sources.

The source of material quantity data for the *material inventory* shall be in accordance with one of the following compliance paths:

6.2.2.1 As-Designed Assessment Requirements.

The *material inventory* shall be established using modeled information in accordance with the *construction documents*.

6.2.2.2 As-Built Assessment Requirements.

The *Bill of Materials* shall be established using measured quantities of materials in accordance with *record documents* or through on-site measurement of material quantity or a combination of both.

6.3 Environmental Impact Data.

The environmental impact data used to represent products or materials shall be consistent with that of the *building* being assessed and the information and scenarios employed in the assessment shall cover all stages of the building's lifecycle.

6.3.1 Allowable Data Sources.

The most representative data source for the products and materials shall be used, as indicated in Table 6.3.2. When the data source most representative of the *product stage* does not include additional lifecycle stages, generic data shall be used for remaining lifecycle stages.

6.3.1.1 Environmental Product Declarations. EPDs used shall be valid, independently verified, Type III, according to ISO Standard 14025 and ISO Standard 21930:2017 or EN Standard 15804 plus Amendment 2. EPDs using EN Standard 50693 shall be accepted for electrical equipment. EPDs not meeting these criteria must include additional data source uncertainty values as documented in Section 6.4.2.

Informative note: EPDs are produced with varying levels of product and supply chain specificity, falling into the following categories:

- a. **Industry-wide EPD.** shall represent the weighted-average environmental performance of an industry's product.
- b. **Product-specific EPD.** Shall reflect manufacturer-specific data.
- c. **Facility-specific and/or supply-chain specific EPD.** shall be a product-specific EPD that uses supply-chain and/or facility-specific data in the LCA process for the chosen product.

6.3.1.2 National or regional datasets of generic environmental impact figures. Shall be compliant with ISO 14040 and 14044 and represent regional supply chains and regionally specific manufacturing practices.

Informative note: This includes datasets such as Federal LCA Commons or BECD Database.

6.3.1.3 Background Datasets. Shall be produced in accordance with ISO 14040 and 14044 following a documented methodology that is consistent and publicly available.

Informative note: This includes data utilized in a publicly available software or database that is produced in a publicly published, consistent underlying methodology.

6.3.1.4 Life Cycle Assessments. ISO Standard 14040 and ISO Standard 14044 provide the foundational LCA standards. Verified LCA reports shall be critically assured to be in compliance with such standards. Unverified LCA results do not incorporate third-party review.

Informative note: CIBSE's TM65 is considered a form of unverified LCA in that it incorporates life cycle thinking but is not compliant with ISO Standard 14040 or ISO Standard 14044 and does not have a process for third-party review.

6.3.2 Data hierarchy and disclosure

6.3.2.1 Data sources

Data sources shall be selected to match the highest appropriate level of resolution, using the following data hierarchy per modeling assessment in Table 6.3.2 below. Uncertainty shall be disclosed and calculated per Section 6.4.2.

Table 6.3.2: Types of environmental impact data to be used during different modeling Assessments

Assessment	Greenhouse Gas Emission Data
<i>As-designed</i>	When a specific product and manufacturer has been selected: a. Facility-specific and/or supply-chain specific <i>EPD</i> (where products/systems are specified) b. Product-specific <i>EPD</i> (where products/systems are specified) c. Industry-wide <i>EPD</i> or regionally-appropriate generic values if no product-specific <i>EPD</i> is available Once materials have been selected, but no specific product and/or manufacturer has been specified or purchased: a. Industry-wide <i>EPD</i> b. Regionally-appropriate generic values from national or regional datasets c. Regionally-appropriate generic values from other background datasets For materials and products, if an <i>EPD</i> is not available: a. Product or Material-specific Life Cycle Assessment b. CIBSE TM 65 product-specific mid-level calculation following local addenda
<i>As-built</i>	When a specific product and manufacturer has been purchased and/or installed: • Facility-specific and/or supply-chain specific <i>EPD</i> (where products/systems are specified) • Product-specific <i>EPD</i> (where products/systems are specified) When a type of product has been purchased and/or installed, but the chosen manufacturer does not have one of the above: • Industry-Wide <i>EPD</i> • Regionally-appropriate generic values from National or Regional datasets • Regionally-appropriate generic values from other background Datasets For materials and products, if an <i>EPD</i> is not available: • Product or Material-specific Life Cycle Assessment • CIBSE TM 65, product-specific mid-level calculation following local addenda

6.4 Quantification of Uncertainty in Embodied GHG Emissions.

An uncertainty factor shall be established per *life cycle stage* based on the uncertainty associated with the quantities in the *Bill of Materials* and with environmental impact data representativeness.

6.4.1 Material Quantity Uncertainty Calculation. This number shall represent the accuracy of the quantities of products and materials within the *bill of materials* based on the data source used for quantity takeoff.

6.4.1.1. As-designed assessment. Where a design phase BIM is used for the *Bill of Materials*, material quantity uncertainty shall be 3%. Where a construction documentation model is used, material quantity uncertainty shall be 1%.

6.4.1.2 As-built assessment. Where an *as-built BIM* is the data source for the *Bill of Materials*, material quantity uncertainty shall be 1%. Where delivered material measurements are used for the *Bill of Materials*, material quantity uncertainty shall be 0%.

6.4.2 Emissions Data Uncertainty Calculation.

For each source of greenhouse gas emissions data, data representativeness must be rated per Normative Appendix B Table B.1 and used to calculate the data uncertainty factor, per Equation 6.4.2.1. This is combined to create a results uncertainty factor per reported value per Equation 6.4.2.2.

Calculation per datum shall follow Equation 6.4.2.1:

$$US = U1 + U2 + U3 + U4 + U5 + U6 \tag{6.4.2.1}$$

Where

- US* is the datum uncertainty score
- U1* is the value of Geographic Representativeness
- U2* is the value of Technological Representativeness
- U3* is the value of Product Representativeness
- U4* is the value of Temporal Representativeness
- U5* is the value of Data Granularity
- U6* is the value of Verifiability.

Based on the value of *US*, the following datum uncertainty factor shall be assigned in accordance with Table 6.4.3.

Table 6.4.3: Data uncertainty factor

<i>US</i>	<10	10 - <15	15 - <20	20 - <25	25 - <30	30 - <35	35 - <40	40
Uncertainty Factor	7%	6%	5%	4%	3%	2%	1%	0%

For each calculation reported from Section 6.5, uncertainty in the emission result should be reported as the sum of the numerical uncertainty factors used within the calculation, per Equation 6.4.2.3:

$$U_R = \sum_{datum}^n UF_{datum} * E_i \tag{6.4.2.3}$$

where

- U_R* is the uncertainty result
- UF_{datum}* is the Uncertainty factor, per Table 6.4.3
- n* is the number of materials or products included in the assessment
- E_i* is the greenhouse gas emissions associated with environmental datum *i*

For final reported uncertainty calculation, this shall be converted back to a percentage figure per Equation 6.4.2.4:

$$U_{RP} = \frac{U_R}{E} \quad (6.4.2.4)$$

Where

U_{RP} is the Uncertainty Result as a percentage

U_R is the result of Equation 6.4.2.3

E is the total reported emissions included in U_R

6.4.3 Reported Uncertainty Calculation.

For each reported value, the total combined uncertainty percentage shall be calculated as the sum of the material quantity uncertainty from Section 6.4.1 and emissions data uncertainty percentage from Equation 6.4.2.4.

6.5 Quantification and Boundaries per *Life Cycle Stages*

6.5.1 Product (Modules A1 through A3)

This stage shall include emissions associated with the creation of finished physical materials to be permanently installed in the *building* and associated *site*. These include all processes associated with raw material extraction, intermediate transportation during the manufacturing process, and manufacturing. Calculation of these emissions follows Equation 6.5.1,

$$EM_{LCS} = \sum_p m_p \times CF_p \quad (6.5.1)$$

where

EM_{LCS} are the *greenhouse gas emissions* for the relevant life cycle stage.

p is a given product or material.

m_p is the quantity of product, p , in accordance with Section 6.2.

CF_p is the product's **embodied carbon factor** for the relevant life cycle stage per Section 5.4.

Informative note: Where the units of m_p and CF_p do not match, the material quantity should be converted to match the units of the **embodied carbon factor** using industry or product-specific density data. Where this data is unavailable, *ASHRAE Handbook of Fundamentals* average density data for the material should be used.

6.5.1.1 As-Designed Compliance Path

Material quantities shall be taken from the *Bill of Materials*, per Section 6.2. The embodied carbon factor for A1-A3 shall be sourced per Section 6.3.3.

6.5.1.2 As-Built Compliance Path:

Material quantities shall be taken from the *Bill of Materials* [Section 6.2] and the embodied carbon factor for A1-A3 for that material or product shall be sourced per Section 6.3.4.

6.5.2 Transportation to Site (Module A4)

This stage shall include all emissions arising from the travel of products from the *point of final manufacture* to the *site*.

6.5.2.1 As-Designed Compliance Path.

Material or product quantities shall be obtained from the material quantities (per Section 6.2) and shall be multiplied by the appropriate emission factor from Table 6.5.2.1.

Exceptions to Section 6.5.2.1:

- 1. For transportation scenarios not covered by Table 6.5.2.1 (e.g. manufactured in other continents) or for construction sites outside of North America, the as-built path should be used based on assumed means of transportation and estimated transportation distances.
- 2. Where A4 is included in the relevant EPD, the corresponding GWP value for A4 shall supersede the transportation factors provided in Table 6.5.2.1.

Table 6.5.2.1: Default emission factors for transportation to site in North America

Type/Source of product	Emission factor	
Ready-mixed concrete from a local plant [1]	11.3	kgCO ₂ e/yd ³
Manufactured products [2]		
Locally manufactured & sourced (within 100 miles)	25.7	kgCO ₂ e/ton
Regionally manufactured & sourced (within 500 miles)	129	kgCO ₂ e/ton
Nationally manufactured & sourced	257	kgCO ₂ e/ton
European manufactured to Eastern US	459	kgCO ₂ e/ton
European manufactured to Western US	331	kgCO ₂ e/ton
Asian manufactured to Eastern US	500	kgCO ₂ e/ton
Asian manufactured to Western US	371	kgCO ₂ e/ton

[1] Established based on data from 2021 NRMCA Fleet Benchmarking and Costs Survey

[2] Established based on EPA Emission Factors for Greenhouse Gas Inventories, 2023. Includes return factor of 0.5 for local and regional trucking.

6.5.2.2 As-Built Compliance Path

Material quantities shall be obtained from the Bill of Materials (per Section 6.2). Transportation distance (d) and applicable means of transportation shall be per contractor tracking or as verified by licensed design professionals.

Calculations of A4 GHG emissions shall follow Equation 6.5.2.2:

$$EM_{A4} = \sum_p m_p \times d_p \times EFO_p \times (1 + RF_p)$$
 (6.5.2.2)

where

p is a given product or material delivered to the site

m_p is the quantity of product or material p expressed as a mass or a volume

d_p is the travel distance of product or material p from last point of sale to the site

EFO_p is the emission factor for outbound trip based on applicable transportation mode from Table 6.5.2.2.

RF_p is the return trip factor for the applicable mode of transportation from Table 6.5.2.2.

Exceptions to 6.5.2.2:

1. In instances where measured data cannot be tracked or verified, the default factors provided in Section 6.5.2. for the *As-Designed* Compliance Path shall be used.
2. Where transport-specific *emissions factors* are known (e.g. *electric vehicles* used for transportation), these shall supersede the transportation factors provided in Table 6.5.2.2.

Table 6.5.2.2: Emission Factors by Transport Mode for Use in Calculating A4 Emissions.

Mode of transportation	GHG emission factor for outbound trip (EFO)	Return trip factor (RF)
Concrete mixing truck	0.381 kgCO ₂ e/yd ³ -mile [1]	1.0
Truck	0.171 kgCO ₂ e per ton-mile [2]	0.5 for trips up to 500 miles 0.0 for trips longer than 500 miles
Air	0.704 kgCO ₂ e per ton-mile [2]	-
Ship or Barge	0.045 kgCO ₂ e per ton-mile [2]	-
Rail	0.021 kgCO ₂ e per ton-mile [2]	-

[1] Established based on data from 2021 NRMCA Fleet Benchmarking and Costs Survey

[2] Established based on EPA Emission Factors for Greenhouse Gas Inventories, 2023.

6.5.3 Construction (A5)

This stage shall include preparation activities occurring on construction site until *project* delivery, inclusive of pre-construction *demolition* (A5-1), if applicable, and construction activities (A5-2), as well the impacts from waste generated during construction (A5-3). Worker transportation to and from the site during the construction process shall be excluded.

6.5.3.1 Pre-construction Demolition (A5-1)

This stage shall include *GHG emissions* due to any *demolition* or *deconstruction* of existing *structures* required for the *project* prior to starting construction, including *GHG emissions* from disposal of materials.

6.5.3.1.1. As-Designed Compliance Path.

GHG emissions shall be estimated based on 35 kg CO₂e/m² of the gross floor area of the *building* being demolished.

6.5.3.1.2 As-Built Compliance Path.

Follow procedure according to Section 6.5.3.2.2 (*As-Built* Compliance Path for A5-2) for *GHG emissions* arising from *demolition* activities on site and Section 6.5.3.3.2 (*As-Built* Compliance Path for A5-3) for *GHG emissions* arising from *demolition* waste.

6.5.3.2 Construction Activities (A5-2)

This stage shall include any activity requiring the use of equipment, tools, or facilities consuming fuel, energy, or water on the construction *site*. This includes, but is not limited to: *site* preparation, groundwork and excavation, ground improvement, connection to utilities, installation of temporary elements necessary for construction, removal of such elements, installation of all materials considered in A1-A4, storage of materials on-site or in the staging area, any field assembly or production occurring on site prior to installation, any transportation or lifting of materials occurring on-site, and use of facilities and trailers by workers on *site*.

6.5.3.2.1 As-Designed Compliance Path.

GHG emissions shall be estimated based on 40 kg CO₂e/m² of the *project* gross floor area.

6.5.3.2.2 As-Built Compliance Path.

Electricity, natural gas, or other fuels, and water consumption shall be determined from utility bills or meter readings. Fuel consumption shall be determined from delivery tickets, invoicing, or estimated from GPS trackers or telematic data. Additional water brought to the *site* shall be determined from delivery tickets, invoices, or submittals. *GHG emissions* associated with energy usage shall be calculated in accordance with Section 7.3.3. *GHG emissions* associated with water usage shall be calculated in accordance with Section 7.4.2.

6.5.3.3 Construction Waste (A5-3)

This stage shall include the manufacturing, transportation, storage, and end-of-life impacts related to materials wasted on *site*, inclusive of packaging. The methodologies for *Modules A1-4, C1-4, and D1-2* [Sections 6.5.1, 6.5.2, 6.5.5, and 6.5.6] in this Standard shall be used to calculate embodied emissions for the quantities of wasted materials.

6.5.3.3.1 As-Designed Compliance Path.

On-site material waste shall be calculated using the wastage rates provided in Table 6.5.3.3, except where otherwise provided in Module A5 data from the relevant *EPD* or in published industry standards. The weight of packaging shall be calculated as 20% of all *construction waste* generated on site, unless packaging is already included in the relevant *EPD*.

Exception to Section 6.5.3.3.1: Packaging for ready-mix concrete shall be taken as zero.

6.5.3.3.2 As-Built Compliance Path.

The same method described in Section 6.5.4.3.3.1 for the *As-Designed* Compliance Path shall be used. Where available, the values from Table 6.5.4.3.3 shall be replaced by *project*-specific wastage rates measured by the contractor responsible for installation. For projects over 2,000 gsf, the following quantities shall be reported by the contractor at completion:

- a. Weight of waste collected for disposal (landfilled or incinerated)
- b. Weight of waste collected to be reused on-site or off-site
- c. Weight of waste collected for recycling (including take-back programs).

770 **Table 6.5.3.3: Default on-site material wastage rates**

	On-site wastage rate (See Note 1)
Pre-fabricated assemblies	
Standardized assembly and small MEP equipment (400 lbs or less) selected from a manufacturer's catalogue. See Notes 2 and 4.	1%
Custom assembly made to order and large MEP equipment (more than 400 lbs). See Notes 3 and 4.	0%
By delivery/installation method	
Delivered by a concrete mixer truck and poured in place (ready-mix concrete)	5%
Sprayed (shotcrete, sprayed fire resistive material)	10%
Soil mixing or jet grouting (grout)	20%
Driven into the ground (piles):	
- precast concrete	5%
- steel piles or casing	3%
- timber piles	20%
Troweled (mortar, skim coating)	15%
Liquid applied (paint, self-leveling concrete topping, roof membranes)	10%
By element size/type	
Blocks and bricks	5%
Tiles (including carpet) and siding	8%
Sheets, boards or panels delivered in standard sizes to be cut as needed on site (See Note 5):	
- metal (steel decking, roofing, flashing, welded wire fabric)	10%
- non-metal (gypsum, plywood)	15%
Elements delivered to site in standard length to be trimmed as needed to required length on site (metal studs, light-weight timber framing, plumbing pipes)	10%
Rebar	3%
Steel connection materials (bolts, studs, welding electrodes, straps, anchors)	5%

771 Notes:

772 1. For materials not listed in this table, the wastage rate shall be taken as 5%.

773 2. Applies to standardized pre-fabricated assemblies or equipment that are shipped to site ready to be installed without any *alterations*
774 resulting in wastage such as cutting or trimming or drilling. For example: standard windows, small appliances.

775 3. Applies to custom pre-fabricated assemblies meeting all of the following: .

776 a. Weighs more than 200 lbs per piece

777 b. Fabricated to the specific dimensions required for the project.

778 c. Shipped to site as a kit of parts or as a single piece ready to be installed without any *alterations* resulting in wastage such as
779 cutting or trimming or drilling.

780 For example: unitized curtainwall, volumetric modular construction, structural steel members, glue-laminated beams and CLT panels,
781 architectural precast panels.

782 4. Any additional material to be applied on site such as coatings or connection materials shall be accounted for separately with applicable
783 on-site wastage ratios.

784 5. Applies to flat products at least 18in in two directions. Otherwise use "tiles" or "elements delivered to site in standard length."

6.5.4 Use Stage (B1)

All material-related *emissions* and carbon removals arising from *building* components during the reference study period shall be reported in *B1*, including: **B1-1**: material emissions and uptake and **B1-2**: *fugitive emissions*.

Informative note: Material emissions included in B1-1 are through processes such as product off-gassing. Uptake is through direct material removal of carbon through processes such as *carbonation* but not including removals from *biomass* growth on the building or site. Detailed guidance on calculating and reporting carbon uptake is given in Section 6.5.13.

6.5.4.1 Material emissions and uptake (B1-1).

Emissions shall be calculated for applicable materials using Equation 6.5.1 from Section 6.5.1. with carbon *emissions factors* for *Module B1*.

6.5.4.2 Fugitive Emissions (Modules B1-2).



6.5.4.2.1 Data Requirements – Fugitive Emissions.

6.5.4.2.1.1 Fugitive Emissions. The *greenhouse gas emissions assessment* shall include *fugitive emissions* occurring during the *installation stage* (Module A5), *use stages* (Modules B1-B5), and *end-of-life stage* (Module C1) of the *project* including refrigerants in refrigeration and HVAC equipment and gases in *permanently installed* fire suppression equipment. A methodology for assessing *greenhouse gas emissions* from *Industrial gases* is included in this section but is not required unless directed by the *authority having jurisdiction*.

6.5.4.2.1.2 Global Warming Potentials. All calculations in this section shall use the 100-year *global warming potentials* from Table C.1 in Normative Appendix C. Where Table C.1 does not include values for gases associated with the *project*, values approved by the *authority having jurisdiction* shall be used.

Exception to 6.5.4.2.1.2: Where the *AHJ* requires a greenhouse gas assessment using a 20-year global warming potential basis and *emissions factors* for all refrigerants and other gases shall use the 20-year global warming potential basis values in Table C.1.

6.5.4.2.1.3 Leakage rates. Leakage rates shall be determined in accordance with Section 6.5.2.3 or Section 6.5.2.4 or the leakage rates from Table C.2 shall be used.

6.5.4.2.2 Quantifying Greenhouse Gas Emissions from Fugitive Emissions

6.5.4.2.2.1 Refrigeration and HVAC Equipment Emissions

6.5.4.2.2.1.1 Equipment and refrigerant inventory. An inventory of refrigeration and HVAC equipment shall be performed and reported in accordance with Section 9. This shall include the types and quantity of equipment, each type of refrigerant used, and mass of each type of refrigerant.

6.5.4.2.2.1.2 Refrigerant replacement. When calculating life cycle emissions for the assessment period, it shall be assumed that the refrigerant will be replaced in kind during the assessment period.

6.5.4.2.2.2 Calculate emissions during installation (Module A5). Determine equipment that was installed during the assessment period and charged on-site. Emissions from equipment that was pre-charged by the manufacturer shall not be included.

6.5.4.2.2.1 As-Designed Compliance Path. Refrigerant leakage rate shall be obtained from Table C.2.

6.5.4.2.2.2 As-Built Compliance Path. Refrigerant leakage rate shall be obtained from contractor submittals where available, with default to Table C.2 where data are unavailable.

For each new piece of equipment, use Equation 6.5.4.2.2.2 to calculate fugitive *greenhouse gas emissions* during installation.

$$GHG_{ref,ie} = \sum_{n=1}^N REF_{ch,site} \times x_i \times GEF_{ref} \quad (6.5.4.2.2.2)$$

where

n = installation event within *building* life-cycle

N = total anticipated installation events throughout the full *building* life-cycle, for each unique refrigerant type and to be estimated based on typical equipment life per ASHRAE Equipment Life Expectancy Chart shown in Appendix A.

$GHG_{ref,ie}$ = Greenhouse gas emissions of leaked refrigerant during installation

$REF_{ch,site}$ = Mass of refrigerant charged within the *site* boundary during installation

x_i = Fraction of refrigerant charged within the *site* boundary and leaked during installation, determined based on the compliance path

GEF_{ref} = Greenhouse gas emissions factor of leaked refrigerant

6.5.4.2.3 Calculate operating emissions (B1). Determine *fugitive greenhouse gas emissions* associated with equipment leaks, piping leaks, and service losses over the assessment period.

6.5.4.2.3.1. As-Designed Compliance Path. Operating refrigerant leakage rate shall be obtained from equipment EPDs where available. Table C.2 shall be used where EPD data are unavailable.

6.5.4.2.3.2 As-Built Compliance Path. An average annual leakage rate for each piece of equipment shall be determined based on the lesser of all available annual refrigerant replacement data and the five most recent years of refrigerant replacement data. For each piece of equipment, use Equation 6.5.4.2.3.2 to calculate emissions.

$$GHG_{ref,oe} = \sum_{t=1}^T REF_{ch,tot} \times x_o \times GEF_{ref} \quad (6.5.4.2.3.2)$$

where

t = timestep

T = Total timesteps of the assessment period

$GHG_{ref,oe}$ = Greenhouse gas emissions of leaked refrigerant during operations

$REF_{ch,tot}$ = Mass of pre-charged refrigerant and refrigerant charged onsite (total refrigerant charge)

x_o = Fraction of total refrigerant charge leaked during operations as determined from Table based on the compliance path

GEF_{ref} = Greenhouse gas emissions factor of leaked refrigerant

6.5.4.2.3.3 Calculate disposal emissions (C4). Identify any disposed equipment during the assessment period.

6.5.4.2.3.3.1 As-Designed and As-Built Compliance Path. The recovery rate used to determine disposal emissions shall be obtained from Table C.2. For each piece of disposed equipment, use Equation 6.5.4.2.3.3.1 to calculate emissions.

$$GHG_{ref,de} = \sum_{n=1}^N REF_{ch,tot} (1 - Z_d) \times GEF_{ref} \quad (6.5.4.2.3.3.1)$$

where

n = disposal event within building life-cycle

N = total disposal events throughout the building life-cycle to be estimated based on typical equipment life per ASHRAE Equipment Life Expectancy Chart in Appendix A.

$GHG_{ref,de}$ = Greenhouse gas emissions of refrigerant at disposal

$REF_{ch,tot}$ = Mass of pre-charged refrigerant and refrigerant charged within the site boundary (total refrigerant charge)

Z_d = Refrigerant recovery efficiency during the disposal process as determined from Table 7.5.1.3

GEF_{ref} = Greenhouse gas emissions factor of leaked refrigerant

6.5.4.2.4 Fire Suppression Operating System Emissions (B1). This section will specify the method to calculate emissions from permanently installed fire suppression systems. Under this method, the user shall multiply the capacity of the system by an emissions factor for the system. Emissions shall be calculated in accordance with section 6.5.4.2.4.1 The user shall perform an inventory of systems and determine the number and types of fire suppression systems, by gas type, and the fire suppressant capacity of each system. If the gas type used has a GWP greater than 0.0, total emissions shall be calculated in accordance with section 6.5.4.2.4.1.

6.5.4.2.4.1 Calculate total emissions. The user shall add the capacities of each fixed/permanently installed system for each gas and multiply the total capacity by the appropriate emissions factor. Annual emissions from fixed/permanently installed systems are assumed to be 2.5 percent (0.025) of the total capacity of the system for each gas. The emission factors provided for this method are as provided in Table B.2. Multiply the emissions of each fire suppressant by its GWP from Table B.2. to calculate CO2 equivalent emissions

$$GHG_{fs} = \sum_{t=1}^T FS_{ch,tot} \times .025 \times GEF_{fs} \quad (6.5.4.2.4.1)$$

Where

GHG_{fs} = Greenhouse gas emissions associated with fixed fire suppression system

$FS_{ch,tot}$ = Total capacity of the fixed installed fire suppression system

.025 = annual fraction of leaked fire suppression gas

GEF_{fs} = Greenhouse gas emissions factor of the fire suppression gas.

6.5.4.2.5 Industrial Gas Emissions

Where the authority having jurisdiction requires that a greenhouse gas assessment include industrial gases, the project shall comply with Sections 6.5.4.2.5.1 and 7.4.2.5.2.

6.5.4.2.5.1 Determine industrial gases. The user shall determine whether any industrial gases with a GWP greater than 0.0 will be used in processes. If so, the user shall estimate the mass of each gas to be purchased

during the assessment period. If data is not available in mass units, the user may need to convert from volume to mass using the density of the specific gas. It is assumed that all gas purchased during the assessment period is used and released during the assessment period. If the user makes a bulk purchase and plans on using the gas over multiple years, the user shall divide the bulk amount by the expected years of usage and consider that to be the purchase amount for the current assessment period, as well as the applicable future assessment periods.

6.5.4.2.5.2. Calculate emissions. The user shall sum the amount of gas to be purchased for each gas type and multiply the total by the appropriate GWP from Table 7.5.1.2(1) or Table 7.5.1.2(2) to calculate CO₂ equivalent emissions.

$$\text{GHG}_{\text{tot}} = (\text{PG1}_{\text{tot}} \times \text{GEF}_{\text{pg1}}) + (\text{PG2}_{\text{tot}} \times \text{GEF}_{\text{pg2}}) \dots \quad (6.5.4.2.5.2)$$

Where

GHG_{tot} = Total *greenhouse gas emissions* from *industrial gases*

PG1_{tot} = Total *industrial gas* type 1 total purchased

GEF_{pg1} = Greenhouse gas emissions factor for *industrial gas* type 1

PG2_{tot} = Total *industrial gas* type 2 total purchased

GEF_{pg2} = *Greenhouse gas emissions* factor for *industrial gas* type 2

6.5.4.2.6 Fire Suppression System Emissions

6.5.4.2.6.1 As-built and operated emissions shall be calculated in accordance with Section 6.5.4.2.4.

6.5.4.2.7 Industrial Gas Emissions

6.5.4.2.7.1 As-built and operated emissions shall be calculated in accordance with Section 6.5.4.2.5 except that actual rather than estimated quantities of gases purchased during the assessment period shall be used.

6.5.5 Maintenance Emissions (B2) shall include all emissions arising from planned activities required for regular validity of warranties, continued efficiency, cleanliness, and good appearance of *building elements*.

6.5.6 Repair Emissions (B3) shall include all emissions associated with the mending, renewal or partial replacement of worn, damaged or degraded components or elements. Emissions estimates for *B3* include emissions from all activities associated with repair, including production of products, transportation to *site*, installation and transportation and disposal of waste.

6.5.6.1 As- Designed Compliance Path: Reserved for future use.

6.5.6.2 As-Built Compliance Path: Reserved for future use.

6.5.7 Replacement Emissions (B4) shall include all emissions associated with the anticipated need to replace *building elements*, including all like-for-like, whole-number replacements of products, components, and *systems* over the *reference study period*. *B4* shall include all emissions arising from production, transportation to *site*, installation, and transportation and disposal of waste of said products.

6.5.7.1 As-Designed Compliance Path: *Building* element service life shall be assumed as depicted in Table A.1. Informative Appendix A Table A.1 Omniclass Building Elements. Where available, product service life shall be

adjusted to match the industry-wide or product-specific *EPD*. Service lives for products contained within larger assemblies shall be adapted to match the expected service life of the assembly for the *project*.

Emissions shall be calculated for applicable material or product replacement by multiplying the expected integer number of material or equipment replacements by the results of Equation 6.5.1 from section 6.5.1.

$$EEEELLLLL \times n = LE \quad (6.5.7.1)$$

where

EEEELLLLL are the *greenhouse gas emissions* for the relevant life cycle stage for a given material or product

n is the integral number of material or equipment replacements (e.g. if the replacement value is 2.3 the integral replacement number is 3 as one will not procure 0.3 of an item)

LE is the resultant lifetime replacement emissions of a given material or product.

Replacement emissions must be considered using the same data for materials and products as was used in *Modules A1–A5* for installation and *Modules C1–C4* for their end of life. Any loads and benefits beyond the *system boundary* from the recovery of materials from *Module A5* or *Modules C1–C4* will also be used for the recovery of any materials in *B4*, reported in *D1*. The treatment of *biogenic carbon* for replacement of biobased materials during the RSP should be in accordance with Section 6.5.13.1.

6.5.7.2 As-Built Compliance Path: Measured service lives shall be used according to maintenance reports or other facility tracking. Emissions shall be calculated for applicable materials and products by multiplying the number of documented material or equipment replacements by the results of Equation 6.5.1 from Section 6.5.1.

Where a new product or assembly is used to replace a previous product or assembly (ex. a new carpet product is used with a different backing and underpad system), the associated *Module B4* emissions shall match the new product.

6.5.8 Refurbishment Emissions (B5). Shall include all emissions associated with *alterations* or improvements to the *building* for it to meet a planned change in function or performance. This shall be considered a new *project*.

Informative note: Refurbishment activities that shall be treated as a new *project* include but are not limited to: major changes to the layout of the interior space of the *building* to meet an anticipated use change or tenant change, the anticipated fit-out of a space that was shelled in the original construction process, the changing of HVAC systems to meet an anticipated change in fuel source or efficiency, a change to the number of stories of a *building*, or a planned extension of the *building* footprint.

6.5.9 Deconstruction and Demolition (C1). Shall include all emissions associated with *deconstruction* or *demolition* of the *project* at the end of its *service life*.

6.5.9.1 As-Designed Compliance Path. Environmental impacts for all site activities associated with C1 shall be included. The environmental impact data shall follow Section 6.3 if known or be based upon 30% of the emissions in A5-2 or 50% of the emissions in A5-2 if the *project* explicitly documents *design for deconstruction* practices or if *deconstruction* is required by the jurisdiction.

6.5.9.2 As-Built Compliance Path. Follow the procedure for A5-1 according to Section 6.5.3.1.2.

6.5.10 Waste Transportation Emissions (C2)

6.5.10.1 Calculations for C2 shall include the transportation of all material from the demolished or deconstructed *project* to the site of disposal or transfer of ownership beyond the *project*. *As-Designed Compliance Path*.

6.5.10.2 Transportation distances shall be based on current waste and recovery facility locations in the *project* Jurisdiction. Material quantities shall be separated based upon their intended disposal pathway and transportation distance shall be calculated per disposal pathway using the relevant facility (e.g. landfill pathway, appropriate *recycling* facility, etc.). Where data are not available for end-of-life disposal practices, emissions shall be calculated for all materials and assemblies using Equation 6.5.1 from Section 6.5.1. with *GHG emissions factors* for C2 from a *harmonized LCA dataset* or *industry-average EPD* as per Section 6.3.1. *As-Built Compliance Path*. Transportation emissions shall be measured according to the requirements for A4, according to Section 6.5.2.2.

6.5.11 Waste Processing for Reuse, Recycling, or Recovery (C3).

This stage shall include emissions associated with material processing for reuse, recovery, and *recycling*. Processing of materials for reuse and recovery shall include collection, basic processing, crushing, chipping, coating and adhesive removal, and bundling.

When materials are combusted for energy recovery, the *GHG emissions* from the activity shall be reported under C3. Combustion without energy recovery shall be reported under C4.

6.5.11.1 As-Designed Compliance Path.

GHG emissions shall be calculated for all materials and assemblies using Equation 6.5.1 from Section 6.5.1. with *GHG emissions factors* for C3 from a *harmonized LCA dataset* or *industry-average EPD* as per Section 6.3.1 or from reported *EPD* as per Section 6.3.1. For bulk materials, waste diversion rates should be used to reflect the jurisdictional practices where available. Where none exist, national averages shall be used.

6.5.11.2 As-Built Compliance Path.

GHG emissions shall be calculated using jurisdiction reports and data to reflect waste treatment practices, or official maintenance manuals outlining waste processing practices from the *project* owner. Where none exist, *GHG emissions* shall be calculated for all materials and assemblies using Equation 6.5.1 from Section 6.5.1. with *GHG emissions factors* for C3 from reported *EPD* data as per Section 6.3.4.

6.5.12 Waste Disposal (C4).

GHG emissions associated with *building elements* sent to landfill shall be reported under C4. This includes all landfill emissions and incineration without energy recovery.

6.5.12.1 As-Designed Compliance Path.

GHG emissions shall be calculated for all materials and assemblies using Equation 6.5.1 from Section 6.5.1. with *GHG emissions factors* for C4 from a *harmonized LCA dataset* or *industry-average EPD* as per Section 6.3.1 or from reported *EPD* data as per Section 6.3.1. For bulk materials, waste diversion rates that reflect the jurisdictional practices shall be used where available. Where none exist, national averages shall be used.

6.5.12.2 As-Built Compliance Path.

GHG emissions shall be calculated using jurisdiction reports and data to reflect waste treatment practices, or official maintenance manuals outlining waste processing practices from the project owner. Where none exist, *GHG emissions* shall be calculated for all materials and assemblies using Equation 6.5.1 from Section 6.5.1 with *GHG emissions factors* for C3 from reported *EPD* data as per Section 6.3.1.

6.5.13 Treatment of Special Topics

6.5.13.1 Treatment of *Carbon Sequestration and Storage*.

Carbon sequestered or stored using any method that is included in the environmental impact data shall be reported separately, noting the relevant life cycle stage(s) of the product when carbon is sequestered.

6.5.13.1.1 Biogenic Carbon.

Biogenic carbon uptake and emissions shall be balanced between nature and the product system for any sustainably sourced timber or biobased products. The sequestered *biogenic* carbon may be reported as a removal of *biogenic* carbon from nature when the *biomass* is harvested (A1-A3) and must be reported as emitted when combusted, decomposed, or degraded at end of life (C3 or C4). Timber or biobased products that are not sustainably sourced shall not include any *biogenic* carbon uptake. Any use of this standard only requiring upfront carbon (A1-A5) shall require a *biogenic* carbon balance where any *biogenic* carbon uptake is reported as stored within the project.

Biogenic carbon stored in building materials shall be reported separately and shall not be combined with fossil emissions.

6.5.13.2 Carbonation.

The *carbonation* process occurs over the life of concrete and cementitious elements and shall be accounted for in the Use (B1; see Section 6.4.4.1) and End-of-Life (C3-4; see Sections 6.4.5.3 and 6.4.5.4) stages where applicable. *Carbonation* may also occur in Module D if recycled concrete aggregate is stockpiled before use. Data from *EPDs* or equivalent sources can be used to account for the impact of *carbonation* in modules B1 and C3 and C4, provided that the conditions in the scenario selected in the data source coincide with the anticipated project-specific ones, particularly in relation to surface area exposure.

6.5.13.3 Delayed Emissions. Reserved for future use.

6.5.13.4 Treatment of Recycled Content.

Carbon impacts associated with recycled material shall be allocated according to the “polluter pays principle”. No burden associated with the original manufacturing of a recycled material shall be allocated to the user of the recycled material.

6.5.13.5 Decarbonization of the Energy Grid or Other Energy During Manufacturing Phases.

To limit the uncertainty in factors impacting assessment results, no projected *decarbonization* shall be included within the minimum required calculations for embodied emissions.

6.5.13.6 Land Use Change.

Both direct and induced *land use change* impacts are considered out of scope under this standard.

7. Quantifying Operational Greenhouse Gas Emissions

7.1 General

7.1.1 Scope. Section 7 specifies requirements for quantifying the operational *greenhouse gas emissions* in the *greenhouse gas emissions* assessment.

7.2 Compliance Paths. The *greenhouse gas emissions* assessment shall comply with Section 7.2.1 and either Section 7.2.2 or Section 7.2.3

7.2.1 Requirements for all Compliance Paths. The *greenhouse gas emissions* assessment shall comply with all of items a. through e. below. Where specified by the *AHJ*, compliance with items f. and g. shall also be required.

- a. Section 7.1, “General”
- b. Section 7.3.1.1, “General Data Requirements – Operational Energy Usage”
- c. Section 7.3.2, “Greenhouse Gas Emissions Factors – Operational Energy Usage”
- d. Section 7.3.3, “Greenhouse Gas Emissions Quantification – Operational Energy Usage”
- e. Section 7.5.1, “Data Requirements – Fugitive Emissions”
- f. Section 7.4.1.1, “General Data Requirements – Operational Water”
- g. Section 7.4.2, “Greenhouse Gas Emissions Quantification – Operational Water”

7.2.2 Requirements for As-Designed Path. Where compliance is shown using the “As-Designed Path,” the *project* shall comply with all the following:

- a. Section 7.3.1.3, “As-Designed Data Requirements – Operational Energy Usage”
- b. Section 7.5.2, “Quantifying Greenhouse Gas Emissions from Fugitive Emissions – As-Designed”

7.2.3 Requirements for As-Built Path. Where compliance is shown using the “As-Built Path,” the *project* shall comply with all the following:

- a. Section 7.3.1.4, “As-Built Data Requirements – Operational Energy Usage”
- b. Section 7.5.3, “Quantifying Greenhouse Gas Emissions from Fugitive Emissions – As-Built”

7.3 Operational Energy Usage (B6)

7.3.1 Operational Energy Usage Data Requirements. The *greenhouse gas emissions* assessment shall comply with Sections 7.3.1.1 and 7.3.1.2 and with either Section 7.3.1.3 or Section 7.3.1.4.

7.3.1.1 Data Time Resolution. *Imported energy* usage shall be determined on an annual basis for each energy source. Where the *AHJ* requires a *greenhouse gas* assessment using time periods other than one year for any *imported energy* source, the energy usage shall be determined for time periods as required by the *authority having jurisdiction* for each applicable *imported energy* source.

7.3.1.2 End Use Data. Energy usage for all of the following shall be determined individually in compliance with Section 7.3.1.1 for the corresponding energy source:

1. All *imported energy*.
2. Electricity usage for *electric vehicle supply equipment* within the *system boundary*.
3. *Imported energy* usage by any *equipment* or *system* that produces *exported energy*.
4. Energy usage for any *equipment*, *system* or end use specified by the *authority having jurisdiction*.

Exception to 7.3.1.2: Where the *project* complies with Section 7.3.1.4 and the end use is not separately metered.

7.3.1.3 As-Designed Data Requirements – Operational Energy Usage. Where the *energy code* applicable to the *project* includes a methodology for a whole building simulated performance-based compliance option, all *imported energy* usage shall be computed in accordance with the requirements for the *proposed design* in that methodology. Where an *energy code* is not applicable or the applicable *energy code* does not have a whole building simulated performance-based compliance option, all *imported energy* usage shall be computed in accordance with the requirements for the *proposed design* in the Performance Rating Method of ANSI/ASHRAE/IES Standard 90.1 for *buildings* other than *low-rise residential buildings*, of ICC IECC Residential for *low-rise residential buildings*, or of ANSI/ASHRAE/IES Standard 90.2 for *residential buildings*.

Exception to 7.3.1.3 Alterations for which the *authority having jurisdiction* approves an alternative computation method for *imported energy* usage and by which no *exported energy* is produced.

7.3.1.4 As-Built Data Requirements – Operational Energy Usage.

7.3.1.4.1 Historical Operational Energy Usage. On-site usage of each *imported energy* source shall be measured and recorded annually and at any additional interval required by the *authority having jurisdiction*.

Exceptions to 7.3.1.4.1

1. Where usage of a *bulk fuel* is not metered, the *authority having jurisdiction* does not require an energy usage interval other than annual for the *bulk fuel*, and the annual usage of the *bulk fuel* is estimated based on the sum of the measured on-site stored fuel at the beginning of a 12-month period and the total documented amount of fuel delivered in the same 12-month period, minus the measured on-site stored fuel at the end of the same 12-month period.
2. Where energy usage measurements include energy usage outside the *system boundary*, the *authority having jurisdiction* approves a method to compute the *imported energy* from the total measurement of the energy source, and the computed values are recorded annually and at any additional interval required by the *authority having jurisdiction*.

7.3.1.4.2 Typical Annual Operational Energy. An average annual energy usage for each *imported energy* source shall be calculated based on the lesser of all available data for the energy source and the five most recent years of energy usage data quantified in accordance with Section 7.3.1.4.1.

Exception to 7.3.1.4.2 Where the *authority having jurisdiction* approves the use of measured *imported energy* usage for a period no less than one year as better representing the future *imported energy* usage.

7.3.2 Greenhouse Gas Emissions Factors – Operational Energy Usage. *Emissions factors* for all *imported energy* shall include:

- a. both *combustion greenhouse gas emissions* and *precombustion greenhouse gas emissions*,

- b. *carbon dioxide* (CO₂), nitrous oxide (N₂O) and methane (CH₄), and
- c. where applicable, electricity transmission and distribution losses.

Where the *authority having jurisdiction* does not require or approve *emissions factors* for an *imported energy* source the *emissions factors* shall comply with Section 7.3.2.1 for *imported energy* sources other than electricity and shall comply with Section 7.3.2.2 for *imported electricity*.

Informative note: See Informative Appendix C for recommended sources for *emissions factors* that may be most appropriate for a particular *jurisdiction* or *project*.

7.3.2.1 Default Greenhouse Gas Emissions Factors – Energy Sources Other Than Electricity.

Emissions factors for all *imported energy* other than electricity shall be taken from Table 7.3.2.1 for a 20-year *global warming potential* basis.

Exceptions to 7.3.2.1:

1. Where the *authority having jurisdiction* requires a *greenhouse gas assessment* using a 100-year *global warming potential* basis and *emissions factors* for all *imported energy* other than electricity are taken from Table 7.3.2.1 for a 100-year *global warming potential* basis.
2. *Emissions factors* for alternative fuels not included in Table 7.3.2.1, approved by the *AHJ*, and calculated in accordance with the California Air Resources Board Low Carbon Fuel Standard or the U.S. Environmental Protection Agency Renewable Fuel Standard.
3. *Emissions factors* for chilled water, steam or hot water approved by the *AHJ* and accounting for all of the following:
 - a. Input fuel and electricity *emissions factors* in accordance with Tables 7.3.2.1, Table 7.3.2.2(1) or Table 7.3.2.2(2).
 - b. Conversion efficiency of the heating or cooling plant.
 - c. Auxiliary equipment and distribution losses associated with delivery of thermal energy to the building.

Table 7.3.2.1 Greenhouse Gas Emissions Factors for Energy Sources Other Than Electricity

Fuel	CO ₂ e(20-year) emissions factor, kg/MWh	CO ₂ e(100-year) emissions factor, kg/MWh
<i>Bulk Fuels</i>		
Fuel Oil (residual)	334	313
Fuel Oil (distillate)	324	303
Liquefied Petroleum Gas or Propane	295	275
Gasoline	337	312
Coal	382	353
Liquefied Natural Gas	330	266
Other <i>bulk fuels</i>	382	353
<i>Energy Utilities</i>		
Natural Gas	277	228
Chilled Water	131	117
Steam	466	383
Hot Water	440	362
Other utility-supplied gases	277	228

Informative Note: Emissions factors in Table 7.3.2.1 include combustion values from the U.S. Environmental Protection Agency and precombustion values from NREL’s “U.S. Life Cycle Inventory Database,” with two exceptions. Precombustion values for natural gas are taken from NETL’s “Life Cycle Analysis of Natural Gas Extraction and Power Generation” (2019). Liquefied Natural Gas includes greenhouse gas emissions associated with liquefaction taken from NETL’s “Life Cycle Greenhouse Gas Perspective on Exporting Liquefied Natural Gas from the United States: 2019 Update” (2019).

7.3.2.2 Default Greenhouse Gas Emissions Factors – Electricity. Emissions factors for electricity shall correspond to the appropriate year of the *reference study period* and the grid region of the *building* or *site* and shall be taken from Table 7.3.2.2(1) for a *greenhouse gas assessment* using a 20-year *global warming potential* basis. Where the year of the *reference study period* (RSP) is before the earliest year in Table 7.3.2.2(1) for the grid region, the value for the earliest year shall be used. Where the year of the RSP is after the latest year in Table 7.3.2.2(1) for the grid region, the value for the latest year shall be used.

Exception to 7.3.2.2 Where the *authority having jurisdiction* requires a *greenhouse gas assessment* using a 100-year *global warming potential* basis, and *emissions factors* for electricity correspond to the appropriate year of the RSP and the grid region of the *building* or *site* in Table 7.3.2.2(2). Where the year of the RSP is before the earliest year in Table 7.3.2.2(2) for the grid region, the value for the earliest year shall be used. Where the year of the RSP is after the latest year in Table 7.3.2.2(2) for the grid region, the value for the latest year shall be used.

Table 7.3.2.2(1) Greenhouse Gas Emissions Factors for Electricity, 20-year Global Warming Potential, kg/MWh

Grid Region	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
AKGD	687.6	693.4														
AKMS	369.0	336.5														
AZNM	515.0	495.9	438.5	381.2	323.9	299.3	274.7	228.5	182.3	167.8	153.2	141.4	129.7	118.0	106.2	94.5
CAMX	376.7	379.1	311.4	243.8	176.1	166.6	157.0	132.3	107.6	87.8	67.9	62.5	57.1	51.6	46.2	40.7
ERCT	528.2	520.1	452.9	385.7	318.6	272.3	226.1	178.4	130.7	110.8	90.9	85.4	79.9	74.4	68.9	63.4
FRCC	586.2	589.5	533.7	477.9	422.0	380.6	339.1	312.7	286.3	264.0	241.7	226.2	210.8	195.4	179.9	164.5
HIMS	875.7	798.5														
HIOA	1085.2	1078.2														
MORE	888.3	921.6	726.2	530.8	335.4	304.4	273.4	239.6	205.8	183.2	160.6	153.6	146.6	139.6	132.6	125.6
MROW	556.2	573.6	491.2	408.8	326.4	271.8	217.2	167.6	118.0	98.9	79.7	75.6	71.4	67.2	63.0	58.8
NEWE	405.7	417.0	341.5	266.0	190.6	188.0	185.4	177.2	169.0	156.9	144.8	139.5	134.2	128.8	123.5	118.2
NWPP	360.6	385.6	339.1	292.5	246.0	216.9	187.8	165.1	142.4	133.2	124.1	118.5	112.8	107.1	101.4	95.8
NYCW	441.3	576.1	412.6	329.8	247.0	211.4	175.9	164.8	153.8	140.9	128.1	122.9	117.8	112.6	107.5	102.3
NYLI	653.4	676.7	610.9	488.3	365.7	313.1	260.4	244.1	227.7	208.7	189.7	182.0	174.4	166.7	159.1	151.5
NYUP	182.3	180.7	170.4	136.2	102.0	87.3	72.6	68.1	63.5	58.2	52.9	50.8	48.6	46.5	44.4	42.3
RFCE	438.8	451.5	417.8	384.1	350.4	338.7	327.0	314.2	301.3	293.0	284.6	276.8	269.1	261.3	253.5	245.7
RFCM	708.6	726.8	681.1	635.4	589.8	549.5	509.2	483.3	457.4	425.4	393.4	389.1	384.8	380.5	376.2	371.9
RFCW	603.2	635.3	622.7	610.2	597.6	578.4	559.2	498.5	437.7	391.0	344.3	331.1	318.0	304.8	291.6	278.4
RMPA	619.6	642.7	648.8	655.0	661.1	585.0	508.9	374.2	239.5	168.2	97.0	93.3	89.7	86.1	82.5	78.8
SPNO	539.2	572.5	536.8	501.1	465.4	357.7	250.1	183.0	115.9	99.1	82.2	79.2	76.2	73.2	70.2	67.2
SPSO	552.5	595.9	444.3	292.8	141.2	118.4	95.5	73.0	50.4	42.6	34.9	32.6	30.4	28.1	25.8	23.6
SRMV	522.0	541.5	536.7	531.8	526.9	507.0	487.1	439.4	391.7	347.1	302.6	283.5	264.4	245.3	226.1	207.0
SRMW	866.7	916.3	834.4	752.5	670.5	638.5	606.4	427.9	249.4	204.9	160.4	154.2	147.9	141.6	135.4	129.1
SRSO	577.8	600.7	550.8	500.9	451.0	426.8	402.5	358.2	313.9	304.4	295.0	281.8	268.6	255.4	242.2	229.0
SRTV	523.2	566.6	605.4	644.1	682.9	638.0	593.0	518.4	443.9	413.6	383.3	363.4	343.4	323.4	303.5	283.5
SRVC	426.0	431.0	401.9	372.8	343.7	323.4	303.0	259.7	216.4	195.0	173.6	160.8	148.0	135.2	122.4	109.6
All Other	520.6	539.1	489.9	440.8	391.6	358.8	326.0	278.4	230.8	207.0	183.2	174.7	166.1	157.5	148.9	140.4

Informative note: Values for Years 2020 and 2021 are computed using the method of *ASHRAE 189.1-2020, Addendum m*, applied to EPA eGRID data. Values for Years 2024, 2026, 2028, 2030, 2035, 2040, 2045 and 2050 are average emissions values taken from NREL's Cambium 2022 Mid-Case Scenario. The EPA eGRID map can be found at: https://www.epa.gov/system/files/images/2023-05/eGRID2021_subregion_map.png. The NREL Cambium map can be found at: <https://www.nrel.gov/docs/fy23osti/84916.pdf>.

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Table 7.3.2.2(1) Greenhouse Gas Emissions Factors for Electricity, 20-year Global Warming Potential, kg/MWh (cont.)

Grid Region	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046	2047	2048	2049	2050
AKGD															
AKMS															
AZNM	87.3	80.2	73.0	65.8	58.6	55.6	52.6	49.6	46.6	43.6	44.8	46.0	47.2	48.4	49.5
CAMX	37.1	33.4	29.8	26.1	22.5	21.2	20.0	18.7	17.5	16.2	17.3	18.3	19.3	20.3	21.4
ERCT	62.0	60.6	59.2	57.7	56.3	56.1	55.8	55.6	55.3	55.1	54.3	53.6	52.8	52.1	51.3
FRCC	161.0	157.5	154.0	150.5	147.0	143.6	140.2	136.8	133.4	129.9	134.6	139.3	144.0	148.7	153.4
HIMS															
HIOA															
MROE	119.4	113.2	107.0	100.9	94.7	100.4	106.2	112.0	117.8	123.5	131.6	139.7	147.8	155.9	164.0
MROW	54.7	50.7	46.6	42.5	38.4	38.1	37.7	37.4	37.0	36.7	35.4	34.2	32.9	31.7	30.4
NEWE	119.1	119.9	120.8	121.7	122.6	125.9	129.1	132.4	135.7	139.0	138.6	138.2	137.8	137.4	137.0
NWPP	91.2	86.6	82.0	77.4	72.8	69.8	66.8	63.9	60.9	57.9	55.7	53.5	51.3	49.1	46.9
NYCW	102.1	101.9	101.7	101.5	101.3	104.0	106.8	109.5	112.3	115.0	118.1	121.3	124.4	127.5	130.6
NYLI	151.2	150.9	150.6	150.3	150.0	154.0	158.1	162.2	166.3	170.3	174.9	179.5	184.1	188.7	193.3
NYUP	42.2	42.1	42.0	41.9	41.8	43.0	44.1	45.2	46.4	47.5	48.8	50.1	51.4	52.6	53.9
RFCE	243.2	240.8	238.3	235.8	233.4	234.7	236.0	237.3	238.6	239.9	242.3	244.7	247.2	249.6	252.1
RFCM	367.1	362.3	357.6	352.8	348.0	343.6	339.2	334.7	330.3	325.9	323.8	321.6	319.5	317.4	315.2
RFCW	260.6	242.7	224.9	207.1	189.2	188.4	187.6	186.8	186.0	185.2	190.2	195.2	200.1	205.1	210.1
RMPA	74.0	69.3	64.5	59.7	54.9	48.3	41.7	35.0	28.4	21.8	22.5	23.3	24.0	24.7	25.5
SPNO	63.3	59.4	55.5	51.6	47.7	43.9	40.0	36.2	32.4	28.5	27.7	26.9	26.1	25.4	24.6
SPSO	22.0	20.4	18.9	17.3	15.8	15.8	15.8	15.8	15.8	15.8	17.3	18.8	20.3	21.8	23.3
SRMV	192.7	178.5	164.2	149.9	135.6	135.8	135.9	136.1	136.3	136.5	145.3	154.2	163.1	171.9	180.8
SRMW	120.7	112.3	103.9	95.4	87.0	82.8	78.5	74.2	69.9	65.7	58.3	50.9	43.6	36.2	28.8
SRSO	217.3	205.6	193.9	182.2	170.6	164.6	158.7	152.7	146.8	140.8	137.3	133.8	130.2	126.7	123.1
SRTV	265.6	247.7	229.8	211.9	194.0	185.9	177.7	169.6	161.5	153.3	152.9	152.5	152.1	151.7	151.3
SRVC	105.7	101.8	97.9	94.1	90.2	81.3	72.5	63.6	54.8	45.9	46.1	46.3	46.4	46.6	46.7
All Other	134.4	128.5	122.5	116.6	110.7	108.4	106.1	103.8	101.5	99.2	99.8	100.3	100.9	101.5	102.1

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Informative note: Values for Years 2020 and 2021 are computed using the method of ASHRAE 189.1-2020, Addendum m, applied to EPA eGRID data. Values for Years 2024, 2026, 2028, 2030, 2035, 2040, 2045 and 2050 are average emissions values taken from NREL's Cambium 2022 Mid-Case Scenario. The EPA eGRID map can be found at: https://www.epa.gov/system/files/images/2023-05/eGRID2021_subregion_map.png. The NREL Cambium map can be found at: <https://www.nrel.gov/docs/fy23osti/84916.pdf>.

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Table 7.3.2.2(2) Greenhouse Gas Emissions Factors for Electricity, 100-year Global Warming Potential, kg/MWh

Grid Region	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035
AKGD	609.2	611.9														
AKMS	334.1	304.9														
AZNM	453.1	436.7	384.3	331.8	279.4	258.5	237.5	197.7	157.9	145.6	133.3	123.0	112.7	102.5	92.2	81.9
CAMX	328.1	329.9	268.3	206.6	144.9	137.1	129.2	108.9	88.6	72.3	55.9	51.4	46.9	42.5	38.0	33.5
ERCT	464.9	459.0	396.7	334.5	272.2	232.1	191.9	151.0	110.0	93.4	76.7	72.1	67.5	62.9	58.3	53.7
FRCC	508.3	511.8	459.0	406.1	353.3	318.6	283.8	261.1	238.3	219.7	201.0	188.0	175.0	161.9	148.9	135.9
HIMS	801.7	731.9														
HIOA	997.4	988.7														
MROE	806.5	837.6	653.2	468.7	284.3	258.0	231.6	202.3	172.9	154.0	135.0	129.1	123.2	117.2	111.3	105.4
MROW	508.8	524.8	447.0	369.1	291.2	242.4	193.5	149.3	105.1	88.2	71.2	67.5	63.7	60.0	56.2	52.5
NEWE	355.1	363.5	294.8	226.0	157.3	155.1	152.9	146.1	139.3	129.4	119.4	115.0	110.6	106.2	101.8	97.4
NWPP	325.0	347.0	303.5	260.1	216.7	190.9	165.0	145.2	125.3	117.4	109.4	104.4	99.4	94.3	89.3	84.3
NYCW	376.8	491.4	350.0	278.6	207.3	176.6	146.0	136.6	127.3	116.7	106.0	101.7	97.3	93.0	88.6	84.2
NYLI	569.8	587.4	529.2	421.3	313.4	267.1	220.7	206.6	192.5	176.4	160.3	153.8	147.2	140.6	134.0	127.4
NYUP	157.8	156.1	146.6	116.7	86.8	74.0	61.1	57.2	53.3	48.9	44.4	42.6	40.8	38.9	37.1	35.3
RFCE	383.4	395.3	361.1	327.0	292.9	282.7	272.5	261.4	250.2	243.3	236.3	229.8	223.4	216.9	210.5	204.0
RFCM	636.0	657.3	610.5	563.6	516.8	477.8	438.7	414.8	390.9	362.1	333.2	328.6	324.0	319.4	314.8	310.2
RFCW	543.6	573.9	556.2	538.5	520.8	503.5	486.1	430.5	374.9	334.4	293.8	282.0	270.2	258.4	246.6	234.8
RMPA	558.9	582.8	584.9	587.0	589.1	521.1	453.1	333.2	213.3	150.0	86.6	83.4	80.2	77.1	73.9	70.7
SPNO	492.2	523.9	488.2	452.5	416.8	320.2	223.5	163.5	103.4	88.5	73.5	70.8	68.1	65.5	62.8	60.1
SPSO	489.7	534.2	396.3	258.3	120.4	100.8	81.1	61.7	42.3	35.9	29.4	27.5	25.6	23.8	21.9	20.0
SRMV	453.6	473.7	465.2	456.7	448.2	430.2	412.2	370.4	328.6	290.9	253.1	237.1	221.1	205.2	189.2	173.2
SRMW	793.0	840.5	760.9	681.3	601.7	572.9	544.1	383.9	223.7	183.9	144.1	138.5	132.9	127.3	121.7	116.1
SRSO	511.3	533.8	485.1	436.5	387.8	367.0	346.2	307.4	268.6	260.6	252.5	241.3	230.1	218.9	207.7	196.5
SRTV	470.6	511.5	542.1	572.7	603.3	563.0	522.7	455.6	388.5	361.8	335.0	317.3	299.6	281.9	264.2	246.5
SRVC	377.3	382.0	353.1	324.1	295.2	277.6	260.0	222.0	183.9	165.5	147.1	136.1	125.1	114.1	103.1	92.1
All Other	463.0	480.9	433.4	386.0	338.6	309.7	280.8	239.0	197.1	176.7	156.3	148.9	141.4	134.0	126.5	119.1

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Informative note: Values for Years 2020 and 2021 are computed using the method of ASHRAE 189.1-2020, Addendum m, applied to EPA eGRID data. Values for Years 2024, 2026, 2028, 2030, 2035, 2040, 2045 and 2050 are average emissions values taken from NREL's Cambium 2022 Mid-Case Scenario. The EPA eGRID map can be found at: https://www.epa.gov/system/files/images/2023-05/eGRID2021_subregion_map.png. The NREL Cambium map can be found at: <https://www.nrel.gov/docs/fy23osti/84916.pdf>.

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Table 7.3.2.2(2) Greenhouse Gas Emissions Factors for Electricity, 100-year Global Warming Potential, kg/MWh (cont.)

Grid Region	2036	2037	2038	2039	2040	2041	2042	2043	2044	2045	2046	2047	2048	2049	2050
AKGD															
AKMS															
AZNM	75.7	69.5	63.3	57.1	50.9	48.3	45.7	43.2	40.6	38.0	39.0	45.0	46.0	47.0	43.0
CAMX	30.5	27.5	24.5	21.5	18.5	17.5	16.5	15.4	14.4	13.4	14.2	19.3	20.1	21.0	17.6
ERCT	52.5	51.3	50.0	48.8	47.6	47.3	47.1	46.8	46.6	46.3	45.7	41.8	41.2	40.5	43.1
FRCC	133.0	130.0	127.1	124.1	121.2	118.4	115.6	112.7	109.9	107.1	111.0	134.3	138.1	142.0	126.5
HIMS															
HIOA															
MROE	100.1	94.8	89.6	84.3	79.0	83.8	88.6	93.4	98.2	103.0	109.7	150.2	156.9	163.7	136.7
MROW	48.9	45.2	41.6	37.9	34.3	34.0	33.6	33.3	32.9	32.6	31.4	24.5	23.3	22.2	26.8
NEWE	98.1	98.8	99.6	100.3	101.0	103.7	106.4	109.0	111.7	114.4	114.1	112.2	111.8	111.5	112.8
NWPP	80.2	76.2	72.1	68.1	64.0	61.3	58.7	56.0	53.4	50.7	48.7	36.7	34.7	32.7	40.7
NYCW	84.0	83.7	83.4	83.2	82.9	85.1	87.4	89.6	91.8	94.1	96.5	111.1	113.6	116.0	106.2
NYLI	127.0	126.6	126.2	125.8	125.4	128.7	132.1	135.5	138.9	142.2	145.9	168.0	171.7	175.4	160.7
NYUP	35.2	35.1	34.9	34.8	34.7	35.7	36.6	37.5	38.5	39.4	40.4	46.5	47.6	48.6	44.5
RFCE	201.8	199.7	197.5	195.4	193.2	194.2	195.2	196.2	197.2	198.2	200.1	211.5	213.4	215.3	207.7
RFCM	306.1	302.0	297.8	293.7	289.6	286.0	282.4	278.8	275.2	271.6	269.2	254.8	252.4	250.0	259.6
RFCW	219.5	204.2	188.9	173.6	158.3	157.6	156.9	156.3	155.6	154.9	158.9	182.6	186.6	190.5	174.7
RMPA	66.4	62.1	57.8	53.5	49.2	43.3	37.3	31.4	25.4	19.5	20.2	24.3	24.9	25.6	22.9
SPNO	56.6	53.2	49.7	46.3	42.8	39.3	35.9	32.4	29.0	25.5	24.7	20.2	19.4	18.7	21.7
SPSO	18.7	17.3	16.0	14.6	13.3	13.3	13.3	13.2	13.2	13.2	14.5	22.4	23.8	25.1	19.8
SRMV	161.2	149.2	137.1	125.1	113.1	113.1	113.2	113.2	113.3	113.3	120.8	165.8	173.3	180.8	150.8
SRMW	108.5	100.9	93.3	85.7	78.1	74.2	70.3	66.3	62.4	58.5	51.8	11.6	4.9	-1.8	25.0
SRSO	186.3	176.1	165.9	155.7	145.5	140.3	135.1	130.0	124.8	119.6	116.4	97.2	94.0	90.8	103.6
SRTV	230.8	215.1	199.4	183.7	168.0	160.9	153.9	146.8	139.8	132.7	132.2	129.2	128.7	128.2	130.2
SRVC	88.8	85.5	82.1	78.8	75.5	68.1	60.8	53.4	46.1	38.7	38.7	39.0	39.0	39.1	38.9
All Other	114.0	108.8	103.7	98.5	93.4	91.4	89.4	87.5	85.5	83.5	83.9	86.0	86.4	86.7	85.3

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Informative note: Values for Years 2020 and 2021 are computed using the method of ASHRAE 189.1-2020, Addendum m, applied to EPA eGRID data. Values for Years 2024, 2026, 2028, 2030, 2035, 2040, 2045 and 2050 are average emissions values taken from NREL's Cambium 2022 Mid-Case Scenario. The EPA eGRID map can be found at: https://www.epa.gov/system/files/images/2023-05/eGRID2021_subregion_map.png. The NREL Cambium map can be found at: <https://www.nrel.gov/docs/fy23osti/84916.pdf>.

7.3.3 Greenhouse Gas Emissions Quantification – Operational Energy Usage. The general form of calculating *greenhouse gas emissions* associated with operational energy usage for the *RSP* shall be in accordance with Equation 7.3.3:

$$GHG_{OE} = \sum_s \sum_{t=1}^T E_{s,t} \times e_{s,l,t} \quad (7.3.3)$$

where

GHG_{OE} = total operational energy greenhouse gas emissions of the *RSP*

t = timestep

T = total timesteps of the *RSP*

$E_{s,t}$ = energy usage for *imported energy* source s , at timestep, t , as calculated in accordance with Section 7.3.1

$e_{s,l,t}$ = *emissions factor* for *imported energy* source, s , at location, l , at timestep, t , as determined in accordance with Section 7.3.2

Where the *authority having jurisdiction* does not specify a required method for calculating *greenhouse gas emissions* associated with operational energy usage, the *project* shall comply with Section 7.3.3.1 or Section 7.3.3.2.

7.3.3.1 As-Designed Path Minimum Quantification Requirements. Where the *project* documents compliance in accordance with the “As-Designed Path,” the *greenhouse gas emissions* associated with operational energy usage for the *RSP* shall be calculated in accordance with Equation 7.3.3.1:

$$GHG_{OE} = Y \times \left(\sum_s A_s \times e_s \right) + \sum_{y=y_1}^{y_Y} A_{elec} \times e_{elec,l,y} \quad (7.3.3.1)$$

where

GHG_{OE} = total operational energy greenhouse gas emissions of the *RSP*, kg

Y = number of years of the *RSP*

A_s = annual energy usage for non-electric *imported energy* source s , as calculated in accordance with Section 7.3.1.3, MWh

e_s = *emissions factor* for non-electric *imported energy* source, s , in accordance with Section 7.3.2, kg/MWh

y_1 = the first year of the *RSP*

y_Y = the last year of the *RSP*

A_{elec} = annual electricity usage, as calculated in accordance with Section 7.3.1.3, MWh

$e_{elec,l,y}$ = greenhouse gas emissions factor for electricity for *project* location, l , and year, y , in accordance with Section 7.3.2, kg/MWh

Exception to 7.3.3.1: *Projects* for which the operational energy usage-related *greenhouse gas emissions* for the *RSP* are calculated in accordance with a method approved by the *authority having jurisdiction*.

7.3.3.2 As-Built Path Minimum Quantification Requirements. Where the *project* documents compliance in accordance with the “As-Built Path”, the *greenhouse gas emissions* associated with operational energy usage for the *RSP* shall be calculated in accordance with Equation 7.3.3.2:

$$GHG_{OE} = \sum_s \sum_{y=y_1}^{y_s} E_{s,y} \times e_s + \sum_s (y_Y - y_s) \times (A_s \times e_s) + \sum_{y=y_1}^{y_{elec}} E_{elec,y} \times e_{elec,l,y} + \sum_{y=y_{elec}+1}^{y_Y} A_{elec} \times e_{elec,l,y} \quad (7.3.3.2)$$

where

GHG_{OE} = total operational energy *greenhouse gas emissions* of the *RSP*, kg
 y_1 = the first year of the *RSP*
 y_s = the most recent year of the *RSP* for which complete energy usage data is available for *imported energy* source, s , as measured in accordance with Section 7.3.1.4.1
 y_Y = the last year of the *RSP*
 $E_{s,y}$ = energy usage for non-electric *imported energy* source s , in year, y , as measured in accordance with Section 7.3.1.4.1
 e_s = *emissions factor* for non-electric *imported energy* source, s , in accordance with Section 7.3.2, kg/MWh
 A_s = annual energy usage for non-electric *imported energy* source s , as calculated in accordance with Section 7.3.1.4.2, MWh
 $E_{elec,y}$ = electricity usage in year, y , as measured in accordance with Section 7.3.1.4.1
 $e_{elec,l,y}$ = *greenhouse gas emissions factor* for electricity for *project* location, l , and year, y , in accordance with Section 7.3.2, kg/MWh
 y_{elec} = the most recent year of the *RSP* for which complete electricity usage data is available, as measured in accordance with Section 7.3.1.4.1
 A_{elec} = annual electricity usage, as calculated in accordance with Section 7.3.1.4.1, MWh

Exception to 7.3.3.2: *Projects* for which the operational energy usage-related *greenhouse gas emissions* for the *RSP* are calculated in accordance with a method approved by the *authority having jurisdiction*.

7.4 Operational Water Usage (B7) Compliance with the requirements of Section 7.4 shall not be required except where specifically required by the *AHJ*.

Informative note: For most buildings, the emissions related to water and wastewater are expected to be very small relative to other emissions. Exceptions may occur where water is supplied by a desalination process, for buildings with exceptionally high water consumption, and for buildings that are part of the water supply or wastewater treatment systems.

7.4.1 Operational Water Data Requirements. The *greenhouse gas emissions assessment* shall comply with Sections 7.4.1.1 and 7.4.1.2 and either Section 7.4.1.3 or Section 7.4.1.4.

7.4.1.1 General Data Requirements – Water usage. All water usage shall be determined on an annual basis. Where the *authority having jurisdiction* requires a *greenhouse gas assessment* using a temporal resolution other than one year for water usage, the water usage shall be determined at the same temporal resolution required by the *authority having jurisdiction*.

7.4.1.2 General Data Requirements – Wastewater generation. All wastewater generation shall be determined on an annual basis. Where the *authority having jurisdiction* requires a *greenhouse gas assessment* using a temporal resolution other than one year for wastewater generation, the wastewater generation shall be determined at the same temporal resolution required by the *authority having jurisdiction*.

7.4.1.3 As-Designed Data Requirements – Operational Water. Water consumption and wastewater generation shall be determined based on the characteristics of the *building* including number of occupants, occupant activity and schedules, water fixture flow rates, and any other relevant factors.

Informative Note: The US DOE provides guidance on estimating water consumption at <https://www.energy.gov/eere/femp/estimating-methods-determining-end-use-water-consumption>.

7.4.1.4 As-Built and Operated Data Requirements – Operational Water. The water usage shall be determined in accordance with Section 7.4.1.4.1. The wastewater generation shall be determined in accordance with Section 7.4.1.4.2.

7.4.1.4.1 As-Built and Operated Data Requirements – Water Usage. Where an *existing building* has not less than one complete year of metered water consumption data, the consumption shall be that of the most recent 12-month period. Where the *building* has water supplied from multiple sources with individual meters (for example, potable water and recycled water), water consumption for each source shall be determined separately.

Exception to 7.4.1.4.1: Where a 12-month period of metered water consumption other than the most recent 12-month period is approved by the *AHJ* as better representing future consumption, consumption data from such period shall be used.

7.4.1.4.2 As-Built and Operated Data Requirements – Wastewater. Where an *existing building* has not less than one complete year of metered wastewater data, the generation shall be that of the most recent 12-month period.

Exception to 7.4.1.4.2: Where a 12-month period of metered wastewater generation other than the most

recent 12-month period is approved by the *AHJ* as better representing future generation, generation data from such period shall be used.

7.4.2 Greenhouse Gas Emissions Quantification – Operational Water. The *greenhouse gas emissions* associated with water usage shall be determined in accordance with Section 7.4.2.1. The *greenhouse gas emissions* associated with wastewater generation shall be determined in accordance with Section 7.4.2.2.

7.4.2.1 Greenhouse Gas Emissions Quantification – Water Usage.

7.4.2.1.1 Water Supply-Related Energy Usage. The energy needed for the supply of water across the analysis boundary shall be determined and shall be expressed as an energy per unit of water volume. The water-related energy consumption shall be the energy per unit water volume multiplied by the water consumption determined in accordance with Section 7.4.1. Where a *building* has multiple water supplies, the energy consumption per unit water volume shall be determined separately for each supply.

The energy per unit water volume shall include energy needed for sourcing, treatment, and transport. Data used shall be provided by the local water utility or as otherwise acceptable to the *AHJ*. Where data for the local water utility are not available, a value of 0.75 kWh/m³ of electricity shall be used. Locations not served by a water utility shall include the energy consumption of local *equipment*, such as well pumps, where such *equipment* is not included in the energy consumption of the *building* as determined in accordance with Section 7.3.

Informative notes:

- Sourcing energy includes well pumps, desalination plants, and any other electric and non-electric energy needed to provide water to a treatment plant delivering water to the site.
- Treatment energy includes the electric and non-electric energy consumption of water treatment plants in the process of filtration and chemical treatment as necessary to make the water potable or otherwise useable on the site.
- Transport energy includes electric and non-electric energy used by pumps necessary to transport the water from the treatment plant to the site including energy needed to provide pressure to the site.

7.4.2.1.2 Water Supply-Related Energy Emissions. Water supply-related *greenhouse gas emissions* shall be calculated in accordance with Section 7.3.3 by replacing the operational energy consumption values with the water supply-related energy usage determined in accordance with Section 7.4.1.

7.4.2.2 Greenhouse Gas Emissions Quantification – Wastewater Generation.

7.4.2.2 Wastewater-Related Energy Usage. The energy needed for the pumping of wastewater from the analysis boundary to a wastewater treatment plant shall be determined and shall be expressed as an energy per unit of water volume. Data used shall be provided by the local wastewater utility or as otherwise acceptable to the *AHJ*. Where data for the local water utility are not available, values of 0.56 kWh/m³ of electricity plus 0.23 kWh/m³ of natural gas shall be used. The wastewater-related energy consumption shall be the energy per unit wastewater volume multiplied by the wastewater generation determined in accordance with Section 7.4.1.

7.4.2.2.2 Wastewater-Related Energy Emissions. Wastewater energy-related *greenhouse gas emissions* shall be calculated in accordance with Section 7.3.3 by replacing the operational energy consumption values with the wastewater energy usage determined in accordance with Section 7.4.1.

7.4.2.2.3 Wastewater-Related Non-Energy Emissions. Non-energy *emissions factor* from wastewater treatment processes shall be determined from the local wastewater treatment system on an emissions per unit of wastewater volume basis. Data used shall be provided by the local wastewater utility or as otherwise acceptable to the *AHJ*. The non-energy wastewater *CO2e* emissions shall be the non-energy *emissions factor* multiplied by the wastewater volume determined in accordance with Section 7.4.1.

7.4.2.2.4 Total Wastewater-Related Emissions. The total wastewater *CO2e emissions* shall be the sum of the wastewater energy related *CO2e emissions* determined in accordance with Section 7.4.2.2.2 and the non-energy wastewater *CO2e emissions* determined in accordance with Section 7.4.2.2.3.

8. Quantifying Greenhouse Gas Emissions Associated with Benefits and Loads Beyond the System Boundary

8.1 General

8.1.1 Scope. Section 8 specifies requirements for quantifying the *greenhouse gas emissions* beyond the *system boundary* in the *greenhouse gas emissions assessment*.

8.2 Compliance Paths. The *greenhouse gas emissions assessment* shall comply with Section 8.2.1 and either Section 8.2.2 or Section 8.2.3.

8.2.1 Requirements for all Compliance Paths. The *greenhouse gas emissions assessment* shall comply with all of the following:

- a. Section 8.1, “General”
- b. Section 8.3, “Recovery, Reuse and Recycling Potential (D1)”
- c. Section 8.4.1.1, “Data Time Resolution”
- d. Section 8.4.1.2, “Specific Exported Energy Sources”

8.2.2 Requirements for As-Designed Compliance Path

- a. Section 8.4.1.3, “As-Designed Data Requirements – Exported Energy”

8.2.3 Requirements for As-Designed Compliance Path

- a. Section 8.4.1.4, “As-Built and Operated Data Requirements – Exported Energy”

8.3 Recovery, Reuse and Recycling Potential (D1). The *greenhouse gas emissions* associated with recovery, reuse and recycling potential of each product in the *material inventory* shall be the appropriate value from the same environmental impact data source as used for Section 6.5.7. Where the environmental impact data source used for the product does not include information on recovery, reuse and recycling potential, the *greenhouse gas emissions* for the product’s recovery, reuse and recycling potential shall be zero.

Exception to 8.3: Where compliance is shown using the “As-Designed Path” and the *construction documents* include all information necessary for *design for deconstruction* of the product or compliance is shown using the “As-Measured Path” and the *record documents* include all information necessary for *design for deconstruction* of the product, and the *greenhouse gas emissions* for the product’s recovery,

reuse and recycling potential are assigned a value equal to the *product stage* calculated in accordance with Section 6.5.1.

8.4 Exported Energy (D2)

8.4.1 Exported Energy Data Requirements. The *greenhouse gas emissions assessment* shall comply with Section 8.4.1.1 and Section 8.4.1.2 and with either Section 8.4.1.3 or Section 8.4.1.4.

8.4.1.1 Data Time Resolution. All *exported energy* shall be computed on an annual basis for each energy source. Where the *authority having jurisdiction* requires a *greenhouse gas assessment* using a temporal resolution other than one year for any *exported energy* source, the *exported energy* shall be computed at the same temporal resolution required by the *authority having jurisdiction* for each applicable *exported energy* source.

8.4.1.2 Specific Exported Energy Sources. All the following specific *exported energy* sources shall be determined individually:

1. Electricity usage for *electric vehicle supply equipment* computed in accordance with Section 7.3.1.2.
2. Any *exported energy* source specified by the *authority having jurisdiction*.

Exception to 8.4.1.2: Where the *project* complies with Section 8.5.1.4 and the specific *exported energy* source is not separately metered.

8.4.1.3 As-Designed Data Requirements – Exported Energy. *Exported energy* shall be computed using the same model used to comply with Section 7.3.1.3.

8.4.1.4 As-Built Data Requirements – Exported Energy. Historical *exported energy* shall be computed in accordance with Section 8.4.1.4.1, and typical annual *exported energy* shall be computed in accordance with Section 8.4.1.4.2 for each *exported energy* source.

Exception to 8.4.1.4: Where an *exported energy* source is not separately metered.

8.4.1.4.1 Historical Exported Energy. Each *exported energy* source shall be measured and recorded annually and at any additional interval required by the *authority having jurisdiction*.

8.4.1.4.2 Typical Annual Exported Energy. An average annual *exported energy* for each *exported energy* source shall be calculated based on the lesser of all available data for the energy source and the five most recent years of *exported energy* data quantified in accordance with Section 8.4.1.4.1.

Exception to 8.4.1.4.2 Where the *authority having jurisdiction* approves the use of measured *exported energy* usage for a period no less than one year as better representing the future *imported energy* usage.

9. Documentation of Greenhouse Gas Emissions

9.1 Compliance. The estimated *as-designed* embodied carbon emissions, operational carbon emissions, and total *greenhouse gas emissions* of the *building* and building site or portion thereof shall be made available to the *authority having jurisdiction*.

The *as built* embodied carbon emissions of the *building* and *building site* or portion thereof shall be made available to the *authority having jurisdiction*, if requested.

9.2 Minimum Project Assessment Identification Parameters. The following *Project Assessment Identification Parameters* shall be collected:

- a. Unique identifier.
- b. Project Name.
- c. Date of Submission.
- d. Project City.
- e. Project Country.
- f. Project Postal Code.
- g. Project Data Has Been Anonymized (yes/no).
- h. Project Units (IP or SI).
- i. Building Gross Floor Area.
- j. Project Site Area.
- k. Year of reporting data.
- l. Project phase at time of assessment.
- m. Embodied Carbon Tool used.
- n. Applicable life cycle stages.
- o. Scope Included in Assessment.
- p. Operational Energy Included (yes/no). If operational energy is included, the referenced calculation methodology shall be stated.
- q. Biogenic Embodied carbon Included (yes/no).

9.3 Object of Assessment.

9.3.1. Physical Limit. Text description that includes the physical limits of the study as defined by Section 5.2.

9.3.2 System Boundary. Text description to identify which building lifecycle stage modules are excluded and why.

9.3.3 Exclusions of Required Processes, Flows and Activities. Text description to identify which *building lifecycle stage* modules are excluded and why.

9.3.4 Reference Study Period. A single value representing the number of years.

9.3.5 Spatial Boundary. Text description to identify unusual or non-standard conditions that are or are not included in the scope of the assessment.

9.4 Embodied Carbon. The *greenhouse gas emissions* associated with embodied carbon shall be documented as the sum of results from the following building element categories as described in Table A1: OmniClass Building Elements (L1-3):

- a. *Substructure*.
- b. *Superstructure (Structure)*.
- c. *Superstructure (Enclosure)*.
- d. Interiors.
- e. Services.
- g. *Special Construction and Demolition*.
- h. Sitework. Elements beyond the minimum required scope for the *building* as noted in the sections above shall be reported independently as *greenhouse gas emissions* for the *building site* and not included within the calculation of the *greenhouse gas emissions for the building*. This shall include documenting Sitework as an independent category separate from other building element categories.
- i. Other voluntarily reported elements beyond the minimum required scope for the *building* shall be reported independently and shall not be included within the calculation of the *greenhouse gas emissions for the building*.
- j. Other voluntarily reported elements beyond the minimum required scope for the *building site* shall be reported independently and shall not be included within the calculation of the *greenhouse gas emissions for the building site*.

9.4.1 Life Cycle Stages. Documentation shall include emissions for the following *life cycle stages* per building element category:

- a. *Product stages A1-A3*.
- b. *Construction stage A4* for the transportation of all materials and products to the construction site.
- c. *Construction stage A5* for the construction processes and installation of materials.
- d. *Use stage B1* for in-use emissions.
- e. *Use stage B4* for the replacement of components in their entirety at the end of their service life.
- f. *Use stage B5* for any refurbishment or retrofit.
- g. *Deconstruction and end-of-life stage C1* for the *deconstruction* and *demolition* of a *project*.
- h. *Deconstruction and end-of-life stage C2* for the transportation of materials to and from intermediate storage or waste processing locations.
- i. *Deconstruction and end-of-life stage C3* for the processing of materials for reuse, recovery, or recycling.
- j. *Deconstruction and end-of-life stage C4* for the treatment and disposal of materials.

9.4.2 Carbon Emissions per Stage. The required documentation of carbon emissions per stage shall include:

- a. GWP Fossil (kgCO₂e).
- b. GWP Biogenic (kgCO₂e).
- c. GWP Captured and stored (kgCO₂e).

9.5 Operational Carbon. The *greenhouse gas emissions* associated with operational carbon shall be documented as the sum of results from the following sources:

- a. Energy consumption in accordance with Section 9.5.1.
- b. Water/Wastewater in accordance with Section 9.5.5.

9.5.1 Greenhouse Gas Emissions from Energy. Documentation for the operational carbon emissions associated with energy shall include the first year and sixty-year projections of *greenhouse gas emissions* from the following sources:

- a. Electricity consumption in accordance with Section 9.5.2.
- b. Fuel consumption in accordance with Section 9.5.3.
- c. Thermal energy consumption in accordance with Section 9.5.4.

9.5.1.1 Global Warming Potential of Energy Sources. The documentation shall state whether the basis of the global warming potential for energy sources is determined on a 20-year or 100-year basis.

9.5.1.2 Energy Model Criteria. The documentation shall include the following key energy model criteria:

- a. A text description naming the energy modeling methodology reference standard used.
- b. The name and version of software used for energy modeling.
- c. The *building* or *building site* location's weather data information based on the information available in ASHRAE's Weather Data Viewer (for details, see <https://www.ashrae.org/technical-resources/bookstore/weather-data-center>). The documentation shall state the weather station name and the World Meteorological Organization (WMO) station identifier, the latitude, longitude, and ASHRAE Climate Zone of the *building* or *building site* per ASHRAE Standard 169, year of accessing the weather file, and hyperlink if file is available in a public domain. The documentation shall state whether the weather file is derived from statistical assessment of historical data or projected weather data files downscaled from global climate models. Projected weather data file documentation shall include source, downscaling methodology, emissions scenarios, projected time period, and associated Climate Model Intercomparison Project (CMIP).

9.5.1.3 Exported Energy Not Applicable. Only consumed energy shall be tracked for compliance with this Standard. Exports of renewable energy to the utility or another user shall not be counted as deductions in the calculation of the greenhouse gas emissions under this standard.

9.5.2 Electricity usage. Documentation of *greenhouse gas emissions* associated with source electricity use shall include:

- a. Name of Electricity Source Provider.
- b. The type of electricity source selected from one of the following: "*imported from a utility*," "*on-site electricity generation*," or "*direct-wired from dedicated offsite source*."
- c. Carbon factor (in kgCO₂e/MWh) stated as either a single annualized value or as a list of sub-annual schedule-based values, with a text description of the source of the data.
- d. Basis of future carbon factor projections if applied to the *RSP* calculations, the basis of such projections shall be documented along with the source of the assumption. Default requirement is the number of years within the *RSP* multiplied by the emissions associated with the first year of operation.
- e. First year *as-designed* consumption (in MWh/year).
- f. First year *as-designed* carbon emissions (in kgCO₂e).
- g. Total life-cycle *as-designed* carbon emissions (in kgCO₂e) based on the *RSP*.

9.5.3 Fuel consumption. *Greenhouse gas emissions* associated with source fuel use shall be documented as follows:

- a. Name of Fuel.
- b. Name of Fuel Source Provider.
- c. Type of Fuel Source. To be selected from "*Piped source*" or "*Bulk-delivery source*."
- d. Carbon factor (in kgCO₂e/MWh). To be stated as a single annualized value, with a text description of the source of the data.
- e. First Year *as-designed* consumption (in MWh/year).
- f. First Year *as-designed* carbon emissions (in kgCO₂e).
- g. Total life-cycle *as-designed* carbon emissions (in kgCO₂e) based on the *RSP*.

9.5.4 Thermal Energy consumption. Documentation of *Greenhouse gas emissions* associated with source thermal energy shall include:

- a. Name of Thermal Energy Source Provider.
- b. The type of Fuel Source for cooling and/or heating, of *conditioned spaces* where cooling refers to chilled water or condenser water and heating refers to heating water or steam.
- c. Carbon factor (in kgCO₂e/MWh of thermal energy) stated as a single annualized value, with a text description of the source of the data.
- d. First Year *as-designed* consumption (in MWh of thermal energy/year).
- e. First Year *as-designed* carbon emissions (in kgCO₂e/year).
- f. Total life-cycle *as-designed* carbon emissions (in kgCO₂e) based on the *RSP*.

9.5.5 Greenhouse Gas Emissions from Water/Wastewater. Documentation of the operational carbon emissions associated with water shall include the first year and sixty-year projections of *greenhouse gas emissions* for the following sources and products:

- a. Potable water.
- b. Reclaimed water.
- c. Wastewater when required by the *Authority Having Jurisdiction*.

9.5.5.1 Water/Wastewater Calculation Criteria. The documentation shall include a text description describing the derivation of the consumption values for each water source or wastewater product and specify:

- a. Name of Water source provider or Wastewater handling utility.
- b. The type of Water/Wastewater. To be selected from potable, reclaimed, or wastewater.
- c. Energy Intensity factor (in kWh/m³) for electricity used in sourcing, treatment, and transport as a single annualized value, and a text description of the source of the data.
- d. Energy Intensity factor (in kWh/m³) for fuel used in sourcing, treatment, and transport as a single annualized value, and a text description of the source of the data.
- e. First year *as-designed* water or wastewater volume (in m³/year).
- f. First year *as-designed* carbon emissions (in kgCO₂e/year) from electricity and fuel use.
- g. Total life-cycle *as-designed* carbon emissions (in kgCO₂e) from electricity and fuel use based on the *RSP*.

9.6 Total Greenhouse Gas Emissions Summary Results. The total *greenhouse gas emissions* for the *building* and *building site* shall be separately stated as follows:

- a. The total *greenhouse gas emissions* is represented by the sum for each *life-cycle stage*, with operational energy and operational water total life-cycle *greenhouse gas emissions* inserted into stages B6 & B7.
- b. For each stage other than B6 & B7, results shall be reported for GWP Fossil Fuel (kgCO₂e), GWP *Biogenic* (kgCO₂e), and area normalized GWP Fossil Fuel (kgCO₂e/m²), GWP *Biogenic* (kgCO₂e/m²) using the building gross floor area or the project site area, respectively for building and site emissions, where areas are as described in Section 5.
- c. For B6 and B7, results shall be reported as GWP Fossil Fuel (kgCO₂e) and area normalized GWP Fossil Fuel (kgCO₂e/m²).
- d. A totalizing sum of Modules A through C for the *building* shall be provided in GWP Fossil Fuel (kgCO₂e), GWP *Biogenic* (kgCO₂e), and area normalized GWP Fossil Fuel (kgCO₂e/m²), GWP *Biogenic* (kgCO₂e/m²) using the building gross floor area.
- e. As appropriate, a totalizing sum of Modules A through C for the *building site* shall be provided in GWP Fossil Fuel (kgCO₂e), GWP *Biogenic* (kgCO₂e), and area normalized GWP Fossil Fuel (kgCO₂e/m²), GWP *Biogenic* (kgCO₂e/m²) using the *building site* area.

9.7 Module D Documentation requirements. The benefits and loads beyond the *system boundary* shall be reported as follows:

- a. *Module D1* (recovery, reuse, and recycling potential) shall be documented (in kgCO₂e) for relevant processes and material flows per *building* element category that occur outside of the *system boundary*.
- b. *Module D2* (exported energy) shall be documented (in MWh/year) for energy serving *electric vehicle supply equipment*.
- c. For each *exported energy* source as specified by the *authority having jurisdiction*, independent documentation of *Module D2* (exported energy) per export type (in MWh/year) shall be provided.

9.8 Environmental Impact Data Disclosure per Product or Material. Documentation of data source per product or material shall include:

- a. Name of Product or Material.
- b. Associated *building* element category
- c. Type of Allowable Data Source. The allowable data source shall be selected from one of the following: “Product-specific *EPD*”, “Facility-specific or supply-chain specific *EPD*”, “Industry-wide *EPD*”, “regional dataset of generic environmental impact figures”, “background datasets”, or “MEP lifecycle assessments” as described under Section 6.3.1. Documentation shall include:
 - d. The publisher’s name.
 - e. The publication date.
 - f. Environmental impact results by life cycle stage for GWP Fossil and GWP *Biogenic* (in kgCO₂e).
 - g. *Functional Unit*.
 - h. Total carbon data uncertainty score as derived from the calculations noted in Section 6.4.

10 NORMATIVE REFERENCES

Reference	Title
ASHRAE	ASHRAE 2020. ANSI/ASHRAE Standard 169, Climatic Data for Building Design Standards. Atlanta: ASHRAE.
ASHRAE	ASHRAE. 2022. ANSI/ASHRAE/IES Standard 90.1, Energy Standard for Buildings Except Low-Rise Residential Buildings, I-P edition. Atlanta: ASHRAE.
ASHRAE	ASHRAE. 2018. ANSI/ASHRAE Standard 209, Energy Simulation Aided Design for Buildings Except Low-Rise Residential Buildings, I-P edition. Atlanta: ASHRAE.
ASHRAE	ASHRAE 2021. Handbook (Fundamentals) , SI edition. Atlanta: ASHRAE
ASHRAE	ASHRAE 2018. ANSI/ASHRAE/IES Standard 90.1, Energy Efficient Design of Low-Rise Residential Buildings, I-P edition. Atlanta: ASHRAE.
ASHRAE	ASHRAE 2021. ASHRAE Weather Data Viewer. https://www.ashrae.org/technical-resources/bookstore/weather-data-center Atlanta: ASHRAE
CIBSE	CIBSE Technical Memorandum 65, Embodied carbon in building services: A calculation methodology (2021)TM65.

CIBSE	CIBSE. (2022). TM65LA Embodied carbon in building services: Using TM65 outside the UK.
CIBSE	CIBSE. (2022). TM65ANZ Embodied carbon in building services: A methodology for ANZ.
CIBSE	CIBSE. (2022). Embodied Carbon Manufacturer Form (Beta 2)
Structural Engineering Institute (SEI) Sustainability Committee	<i>Calculation Methodology for Structural Systems in Whole-Building Life Cycle Assessment</i> , a publication of the Structural Engineering Institute (SEI) Sustainability Committee.
Intergovernmental Panel on Climate Change (IPCC)	Intergovernmental Panel on Climate Change (IPCC) Fourth Assessment Report (2007)
EPA	http://www.epa.gov/ozone/snap/refrigerants/refblend.html
	Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990- 2021
ICC	ICC 2021. International Energy Conservation Code. Washington D.C.: ICC.
ISO	International Organization Standardization [ISO 2020] ISO 14025:2020 – Environment labels and declarations.
ISO	International Organization Standardization [ISO 2020] ISO 14040:2006 Environmental management Life cycle assessment Principles and framework.
ISO	International Organization Standardization [ISO 2017] ISO 21930:2017, Sustainability in buildings and civile engineering works- Core rules for environmental product declarations of construction products and services
EN	EN 2012. EN Standard 15804-2012, Sustainability of construction works, Environmental declarations, Core rules for the product category of construction products plus amendment 2

Informative note: The most recent versions of reference TM 65 documents for the UK, Australia and New Zealand, and the use of TM 65 methodologies outside the UK can be found at <https://www.cibse.org/tm65>, with reference documents at time of publishing as follows:

- CIBSE. (2021). TM65 Embodied carbon in building services: A calculation methodology.
- CIBSE. (2022). TM65LA Embodied carbon in building services: Using TM65 outside the UK.
- CIBSE. (2022). TM65ANZ Embodied carbon in building services: A methodology for ANZ.
- CIBSE. (2022). Embodied Carbon Manufacturer Form (Beta 2)]k

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objectors on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX A

A.1 Building Elements

All *buildings* are made up of discreet *building elements* which can be grouped in consistent categories such as structure, envelope, interiors, and services. A Whole Life Cycle Analysis report format will follow a common elemental *structure* based on **OmniClass Table 21 – Building Elements**. OmniClass is designed to provide a standardized basis for classifying information created and used by the North American architectural, engineering, and construction (AEC) industry, and can be used throughout the full facility and asset life cycle.

All *building elements* referenced in the table below that are within the scope of the *building project* must be included in any compliant WLCA, whatever the *project* phase. While the granularity of reporting and modeling may increase as a design develops with the greatest level of detail and certainty about material estimates after project completion. Increasing the specificity of model data should improve the accuracy of the WLCA and reduce uncertainty, however, it should not result in increased scope.

Table A.1 OmniClass Building Elements (L1-3)

				Years
OmniClass Number	Element Category	Required	Exclude	Default Service Life
21-01 00 00	Substructure			
21-01 10	Foundations	X		Life of building
21-01 10 10	– Standard Foundations	X		Life of building
21-01 10 20	– Special Foundations	X		Life of building
21-01 20	Subgrade Enclosures	X		Life of building
21-01 20 10	– Walls for Subgrade Enclosures	X		Life of building
21-01 40	Slabs-On-Grade	X		Life of building
21-01 40 10	– Standard Slabs-on-Grade	X		Life of building
21-01 40 20	– Structural Slabs-on-Grade	X		Life of building
21-01 40 30	– Slab Trenches	X		Life of building
21-01 40 40	– Pits and Bases	X		Life of building
21-01 40 90	– Slab-On-Grade Supplementary Components	X		Life of building
21-01 60	Water and Gas Mitigation	X		
21-01 60 10	– Building Subdrainage	X		
21-01 60 20	– Off-Gassing Mitigation		X	n/a
21-01 90	– Substructure Related Activities		X	n/a
21-01 90 10	– Substructure Excavation		X	n/a
21-01 90 20	– Construction Dewatering		X	n/a

21-01 90 30	– Excavation Support		X	n/a
21-01 90 40	– Soil Treatment		X	n/a
21-02 10	Superstructure			Life of building
21-02 10 10	– Floor Construction	X		Life of building
21-02 10 20	– Roof Construction	X		Life of building
21-02 10 80	– Stairs	X		Life of building
	– Enclosure			
21-02 20	– Exterior Vertical Enclosures	X		35
21-02 20 10	– Exterior Walls	X		35
21-02 20 20	– Exterior Windows	X		30
21-02 20 50	– Exterior Doors and Grilles	X		30
21-02 20 70	– Exterior Louvers and Vents	X		30
21-02 20 80	– Exterior Wall Appurtenances	X		30
21-02 20 90	– Exterior Wall Specialties	X		30
21-02 30	– Exterior Horizontal Enclosures	X		
21-02 30 10	– Roofing	X		20
21-02 30 20	– Roof Appurtenances		X	n/a
21-02 30 40	– Traffic Bearing Horizontal Enclosures	X		20
21-02 30 60	– Horizontal Openings	X		20
21-02 30 80	– Overhead Exterior Enclosures	X		20
21-03 00 00	– Interiors			
21-03 10	– Interior Construction	X		
21-03 10 10	– Interior Partitions	X		25
21-03 10 20	– Interior Windows	X		25
21-03 10 30	– Interior Doors	X		25
21-03 10 40	– Interior Grilles and Gates		X	n/a
21-03 10 60	– Raised Floor Construction	X		25
21-03 10 70	– Suspended Ceiling Construction	X		(Refer to Interior Finishes below; 21-03 20 50)
21-03 10 90	– Interior Specialties		X	n/a
21-03 20	– Interior Finishes	X		
21-03 20 10	– Wall Finishes	X		
	- Epoxy (Two part)			15
	- Fabric			5
	- Glazed Wall Tile			50
	- Paint			5
	- Plastic Laminate			10

	- Vinyl Wall Covering			10
	- Wall base			5
	- Wallpaper			4
	- Wood Finishes			
	- Finish			10
	- Substrate			50
21-03 20 20	— Interior Fabrications	X		12
21-03 20 30	— Flooring	X		
	- Carpet			
	- Broad Loom			5
	- Carpet Tiles			5
	- Loop Pile			15
	- Concrete			50
	- Epoxy Coating (Two Part)			10
	- Hardwood			
	- Finish			10
	- Substrate			50
	- Laminate			10
	- Linoleum			5
	- Mosaic Tile			50
	- Porcelain Tile			50
	- Rubber Tile			5
	- Stone			
	- Granite			75
	- Marble			50
	- Terrazzo			50
	- Vinyl			
	- Sheet			12
	- Tile			12
21-03 20 40	— Stair Finishes	X		See flooring.
21-03 20 50	— Ceiling Finishes	X		
	- Drywall / Plaster			30
	- Metal			25
	- Suspended			
	- Ceiling Tiles			13
	- Lay-in systems			25
	- Spline Systems			20
21-04	— Services			
21-04 10	— Conveying		X	

21-04 10 10	– Vertical Conveying Systems		X	15
21-04 10 30	– Horizontal Conveying		X	20
21-04 10 50	– Material Handling		X	n/a
21-04 10 80	– Operable Access Systems		X	n/a
21-04 20	– Plumbing	X		
21-04 20 10	– Domestic Water Distribution	X		
	- Copper Pipe			30
	- Pex Water Pipe			30
	- PVC / CPVC Pipe			30
	- Stainless Steel Pipe			30
21-04 20 20	– Sanitary Drainage	X		
	- Cast Iron Pipe			30
	- PVC / CPVC Pipe			30
21-04 20 30	– Building Support Plumbing Systems		X	
21-04 20 50	– General Service Compressed-Air		X	
21-04 20 60	– Process Support Plumbing Systems		X	
21-04 30	– Heating, Ventilation, and Air Conditioning (HVAC)	X		
21-04 30 10	– Facility Fuel Systems	X		
21-04 30 20	– Heating Systems	X		
	- Heat Pump			15
	- Ground Source Heat Pump			40
	- Boiler			15
21-04 30 30	– Cooling Systems	X		
	- Air Conditioner			10
	- Heat Pump			15
	- Ground Source Heat Pump			40
	- Cooling Tower			15
	- Chiller			15
21-04 30 50	– Facility HVAC Distribution Systems	X		
	- Flexible Ductwork			10
	- Metal Ductwork			20
	- Pumps			10
	- Variable Refrigerant Flow (VRF)			15
	- Variable Air Volume (VAV) Air Handling Unit (AHU) or single duct			15
	- VAV Terminals with electric or fan coil			15
21-04 30 60	– Ventilation	X		
	- Dedicated Outside Air with Electric Heat			15
	- Dedicated Outside Air with Heat Recovery			15
21-04 30 70	– Special Purpose HVAC Systems	X		

21-04 40	– Fire Protection		X	
21-04 40 10	– Fire Suppression		X	
21-04 40 30	– Fire Protection Specialties		X	n/a
21-04 50	– Electrical	X		
21-04 50 10	– Facility Power Generation	X		
	- Backup Generator			20
	- Photovoltaic Collector Panels			20
21-04 50 20	– Electrical Service and Distribution	X		
	- Switchgear			30
	- Transformer			30
21-04 50 30	– General Purpose Electrical Power	X		
	- Battery Backup			5
	- Uninterruptable Power Supply			10
21-04 50 40	– Lighting	X		
	- Light Fixtures			20
21-04 50 80	– Miscellaneous Electrical Systems		X	
21-04 60	– Communications		X	
21-04 60 10	– Data Communications		X	
21-04 60 20	– Voice Communications		X	
21-04 60 30	– Audio-Video Communication		X	
21-04 60 60	– Distributed Communications and Monitoring		X	n/a
21-04 60 90	– Communications Supplementary Components		X	n/a
21-04 70	– Electronic Safety and Security		X	n/a
21-04 70 10	– Access Control and Intrusion Detection		X	n/a
21-04 70 30	– Electronic Surveillance		X	n/a
21-04 70 50	– Detection and Alarm		X	n/a
21-04 70 70	– Electronic Monitoring and Control		X	n/a
21-04 70 90	– Electronic Safety and Security Supplementary Components		X	n/a
21-04 80	– Integrated Automation		X	
21-04 80 10	– Integrated Automation Facility Controls		X	
21-05	– Equipment and Furnishings			
21-05 10	– Equipment		X	n/a
21-05 10 10	– Vehicle and Pedestrian Equipment		X	n/a
21-05 10 30	– Commercial Equipment		X	n/a
21-05 10 40	– Institutional Equipment		X	n/a
21-05 10 60	– Residential Equipment		X	n/a
21-05 10 70	– Entertainment and Recreational Equipment		X	n/a
21-05 10 90	– Other Equipment		X	n/a
21-05 20	– Furnishings		X	n/a

21-05 20 10	– Fixed Furnishings		X	n/a
21-05 20 50	– Movable Furnishings		X	n/a
21-06	– Special Construction and Demolition			
21-06 10	– Special Construction	X		
21-06 10 10	– Integrated Construction	X		
21-06 10 20	– Special Structures	X		
21-06 10 30	– Special Function Construction	X		
21-06 10 50	– Special Facility Components	X		
21-06 10 60	– Athletic and Recreational Special Construction	X		
21-06 10 80	– Special Instrumentation		X	n/a
21-06 20	– Facility Remediation		X	n/a
21-06 20 10	– Hazardous Materials Remediation		X	n/a
21-06 30	– Demolition	X		
21-06 30 10	– Structure <i>Demolition</i>	X		
21-06 30 30	– Selective <i>Demolition</i>	X		
21-06 30 50	– Structure Moving	X		
21-07	– Sitework			
21-07 10	– Site Preparation	X		
21-07 10 10	– Site Clearing	X		
21-07 10 20	– Site Elements <i>Demolition</i>	X		
21-07 10 30	– Site Element Relocations	X		
21-07 10 50	– Site Remediation		X	
21-07 10 70	– Site Earthwork	X		
21-07 20	– Site Improvements	X		
21-07 20 10	– Roadways	X		
21-07 20 20	– Parking Lots	X		
21-07 20 30	– Pedestrian Plazas and Walkways	X		
21-07 20 40	– Airfields	X		
21-07 20 50	– Athletic, Recreational, and Playfield Areas	X		
21-07 20 60	– Site Development		X	
21-07 20 80	– Landscaping		X	n/a
21-07 30	– Liquid and Gas Site Utilities	X		
21-07 30 10	– Water Utilities	X		
21-07 30 20	– Sanitary Sewerage Utilities	X		
21-07 30 30	– Storm Drainage Utilities	X		
21-07 30 50	– Site Energy Distribution	X		
21-07 30 60	– Site Fuel Distribution	X		
21-07 30 90	– Liquid and Gas Site Utilities Supplementary Components		X	
21-07 40	– Electrical Site Improvements	X		

21-07 40 10	– Site Electric Distribution Systems	X		
21-07 40 50	– Site Lighting	X		
21-07 50	– Site Communications		X	
21-07 50 10	– Site Communications Systems			
21-07 90	– Miscellaneous Site Construction	X		
21-07 90 10	– Tunnels	X		

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(This is a normative appendix and is part of the standard.)

Normative Appendix B Data Representativeness

Table B.1: Data representativeness evaluation table

Quality	Geographic Representativeness	Technological Representativeness	Product Representativeness	Temporal Representativeness	Data Granularity	Verifiability
Very Good	Region is representative (e.g. NYC and Northeast US) - 10	Reflects the identical technology and characteristics (e.g. BF/BOF steel with 40% recycled content) - 6	Data for actual product used - 6	Year of data collection less than 3 years from year of construction - 8	Site-specific data (factory-specific) - 5	Third-party verified to EN 15804 - 5
Good	Region is geographically similar (e.g. NYC and Quebec) - 7	Partially reflects technology and technical characteristics (e.g. BOF steel with incorrect recycled content) - 4	Representative or average product adapted by relevant characteristic (e.g. data per m ² adapted by scaling factor) - 5	<5 years from year of construction - 6	Manufacturer-specific (actual manufacturer used and averaged across factories or different factory site) - 4	Independently verified to ISO 14025 - 4
Fair	Much larger region or global dataset (e.g. NYC and NAM) - 4	Does not reflect specifics of the technology (e.g. steel generally) - 3	Average data for product group of actual product used	<10 years from year of construction - 4	Regional Sector Data - 3	Peer reviewed to ISO 14044 - 3
Poor	Totally different context (e.g. China for EU) - 2	Totally different technology (EAF steel for BOF steel) - 1	Proxy data or data extrapolated from a group of datasets - 2	Less than 15 years from year of construction - 2	Global sector data - 2	Not peer reviewed or not verified - 0
Not Accepted				>15 years from construction		

Based on the value of *US*, the following datum uncertainty factor shall be assigned in accordance with Table 6.4.3.

(This is a normative appendix and is part of the standard.)

Normative Appendix C Fugitive Emissions Data

Table C1: Global Warming Potentials for Refrigerants and other gases

Common Name	Formula	GWP-20	GWP-100
Carbon Dioxide	CO ₂	1	1
Methane	CH ₄	81.2	25 (27.9)
Nitrous Oxide	N ₂ O	273	298 (273)
Sulfur hexafluoride	SF ₆		22,800
Nitrogen trifluoride	NF ₃		17,200
HFC-23	CHF ₃		14,800
HFC-32	CH ₂ F ₂		675
HFC-41	CH ₃ F		92
Chlorofluorocarbons			
CFC-11	CCl ₃ F	8320	6230
CFC-12	CCl ₂ F ₂	12700	12500
CFC-13	CClF ₃	12400	16200
CFC-112	CCl ₂ FCCl ₂ F	5620	4620
CFC-112a	CCl ₃ CClF ₂	4740	3550
CFC-113	CCl ₂ FCClF ₂	6860	6520
CFC-113a	CCl ₃ CF ₃	5110	3930
CFC-114	CClF ₂ CClF ₂	8260	9430
CFC-114a	CCl ₂ FCF ₃	7510	7420
CFC-115	CClF ₂ CF ₃	7410	9600
E-R316c	trans cyc (-CClF ₂ CF ₂ CClF-)	4810	4320
Z-R316c	cis cyc (-CClF ₂ CF ₂ CClF-)	5590	5660
CFC 1112	CClF=CClF	0.454	0.126
CFC 1112a	CCl ₂ =CF ₂	0.076	0.021
Hydrochlorofluorocarbons			
HCFC-21	CHCl ₂ F	575	160
HCFC-22	CHClF ₂	5690	1960
HCFC-31	CH ₂ ClF	286	79.4
HCFC-121	CHCl ₂ CCl ₂ F	210	58.3
HCFC-122	CHCl ₂ CClF ₂	203	56.4
HCFC-122a	CHClFCCl ₂ F	879	245
HCFC-123	CHCl ₂ CF ₃	325	90.4
HCFC-123a	CHClFCClF ₂	1410	395
HCFC-124	CHClF ₂ CF ₃	2070	597
HCFC-124a	CHF ₂ CClF ₂	5110	2070
HCFC-132	CHClFCHClF	440	122
HCFC-132a	CHCl ₂ CHF ₂	253	70.4
HCFC-132c	CH ₂ FCCl ₂ F	1220	342
HCFC-133a	CH ₂ ClCF ₃	1370	388

HCFC-141	CH ₂ ClCHClF	168	46.6
HCFC-141b	CH ₃ CCl ₂ F	2710	860
HCFC-142b	CH ₃ CClF ₂	5510	2300
HCFC-225ca	CHCl ₂ CF ₂ CF ₃	491	137
HCFC-225cb	CHClFCF ₂ CClF ₂	1960	568
HCFO-1233zd(E)	(E)-CF ₃ CH=CHCl	14	3.88
HCFO-1233zd(Z)	(Z)-CF ₃ CH=CHCl	1.64	0.454
(e)-1-chloro-2-fluoroethene	(E/Z)-CHCl=CHF	0.13	0.004
Hydrofluorocarbons			
HFC-23	CHF ₃	12400	14600
HFC-32	CH ₂ F ₂	2690	771
HFC-41	CH ₃ F	485	135
HFC-125	CHF ₂ CF ₃	6740	3740
HFC-134	CHF ₂ CHF ₂	3900	1260
HFC-134a	CH ₂ FCF ₃	4140	1530
HFC-143	CH ₂ FCHF ₂	1300	364
HFC-143a	CH ₃ CF ₃	7840	5810
HFC-152	CH ₂ FCH ₂ F	77.6	21.5
HFC-152a	CH ₃ CHF ₂	591	164
HFC-161	CH ₃ CH ₂ F	17.4	4.84
HFC-227ca	CF ₃ CF ₂ CHF ₂	5370	2980
HFC-227ea	CF ₃ CHFCF ₃	5850	3600
HFC-236cb	CH ₂ FCF ₂ CF ₃	3750	1350
HFC-236ea	CHF ₂ CHFCF ₃	4420	1500
HFC-236fa	CF ₃ CH ₂ CF ₃	7450	8690
HFC-245ca	CH ₂ FCF ₂ CHF ₂	2680	787
HFC-245cb	CF ₃ CF ₂ CH ₃	6970	4550
HFC-245ea	CHF ₂ CHFCHF ₂	912	255
HFC-245eb	CH ₂ FCHFCF ₃	1160	325
HFC-245fa	CHF ₂ CH ₂ CF ₃	3170	962
HFC-263fb	CH ₃ CH ₂ CF ₃	269	74.8
HFC-272ca	CH ₃ CF ₂ CH ₃	1910	599
HFC-329p	CHF ₂ CF ₂ CF ₂ CF ₃	5010	2890
HFC-365mfc	CH ₃ CF ₂ CH ₂ CF ₃	2920	914
HFC-43-10mee	CF ₃ CHFCHFCF ₂ CF ₃	3960	1600
HFO-1123	CHF=CF ₂	0.017	0.005
HFO-1132a	CH ₂ =CF ₂	0.189	0.052
HFO-1141	CH ₂ =CHF	0.088	0.024
HFO-1225ye(Z)	(Z)-CF ₃ CF=CHF	1.24	0.344
HFO-1225ye(E)	(E)-CF ₃ CF=CHF	0.426	0.118
HFO-1234ze(Z)	(Z)-CF ₃ CH=CHF	1.13	0.315
HFO-1234ze(E)	(E)-CF ₃ CH=CHF	4.94	1.37
HFO-1234yf	CF ₃ CF=CH ₂	1.81	0.501
HFO-1336mzz(E)	(E)-CF ₃ CH=CHCF ₃	64.3	17.9

HFO-1336mzz(Z)	(Z)-CF ₃ CH=CHCF ₃	7.48	2.08
HFO-1243zf	CF ₃ CH=CH ₂	0.94	0.261
HFO-1345zfc	CF ₃ CF ₂ CH=CH ₂	0.656	0.182
3,3,4,4,5,5,6,6, 6-nonafluorohex- 1-ene	n-C ₄ F ₉ CH=CH ₂	0.734	0.204

- a. Smith, C., Z.R.J. Nicholls, K. Armour, W. Collins, P. Forster, M. Meinshausen, M.D. Palmer, and M. Watanabe, 2021: The Earth's Energy Budget, Climate Feedbacks, and Climate Sensitivity Supplementary Material. In Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change [Masson-Delmotte, V., P. Zhai, A. Pirani, S.L. Connors, C. Péan, S. Berger, N. Caud, Y. Chen, L. Goldfarb, M.I. Gomis, M. Huang, K. Leitzell, E. Lonnoy, J.B.R. Matthews, T.K. Maycock, T. Waterfield, O. Yelekçi, R. Yu, and B. Zhou (eds.)]. Available from <https://www.ipcc.ch/>
- b. For gases not shown in this table, please refer to the tables as found in "The Earth's Energy Budget, Climate Feedbacks and Climate Sensitivity Supplementary Material" at:
https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Chapter07_SM.pdf

Table C.2: Default Leakage Rates for Refrigeration/Air Conditioning Equipment

Type of Equipment	Installation Leakage Rate	Annual Operating Leakage Rate	Recovery Efficiency
Room AC-window/wall/portable	NA	2.0%	80%
Residential AC/HP (ducted)	1.0%	5.0%	80%
Small self contained AC/HP (PTAC/HP)	NA	1.0%	80%
Residential mini-split and multi-split ductless AC/HP	1.0%	2.5%	80%
Commercial ductless VRF Systems	1.0%	3.0%	80%
Light Commercial Rooftop AC/HP units (5-25 tons)	1.0%	8.0%	80%
Large Commercial Rooftop AC/HP units (>25 tons)	1.0%	6.0%	80%
Water-cooled Centrifugal & Screw Chillers (all sizes)	1.0%	3.0%	80%
Air-cooled Chillers (all sizes)	1.0%	6.0%	80%
Residential refrigerators & freezers	1.0%	0.5%	70%
Industrial Refrigeration including Food Processing and Cold Storage	3.0%	25.0%	90%
Commercial refrigeration (all sizes)	3.0%	25.0%	70%
Self-contained refrigeration display cases	3.0%	2.5%	70%
Multiplex Rack Systems	3.0%	30.0%	70%
Remote Condensing Unit Systems	3.0%	15.0%	70%

(This appendix is not part of this standard. It is merely informative and does not contain requirements necessary for conformance to the standard. It has not been processed according to the ANSI requirements for a standard and may contain material that has not been subject to public review or a consensus process. Unresolved objectors on informative material are not offered the right to appeal at ASHRAE or ANSI.)

INFORMATIVE APPENDIX D Informative References

This informative appendix presents a list of available standards and resources that provide *greenhouse gas emission* factors that could be used in place of the default values provided in this standard. It is recommended that these resources are reviewed in more detail to best determine which resource and which emissions factor(s) within them are most appropriate for the proposed application:

International Green Construction Code powered by ANSI/ASHRAE/ICC/USGBC/IES Standard 189.1

Standard for the Design of High-Performance Green Buildings

ANSI/ASHRAE Standard 105 Standard Methods of Determining, Expressing and Comparing Building Energy Performance and Greenhouse Gas Emissions

ANSI/ASHRAE Standard 228 Standard Method of Evaluating Zero Net Energy and Zero Net Carbon Building Performance

EFDB IPCC – Emission factor database, Intergovernmental Panel on Climate Change

IEA - Emission Factors, Annual GHG emission factors for World countries from electricity and heat generation, International Energy Agency

JRC CoM - Greenhouse gases emission factors for local emission inventories, Covenant of mayors for Climate and Energy initiative, CoM databases, Joint Research Center, European Commission